EEE1001
ELECTRIC CIRCUITS AND SYSTEMS

Module 3 – Lecture 1

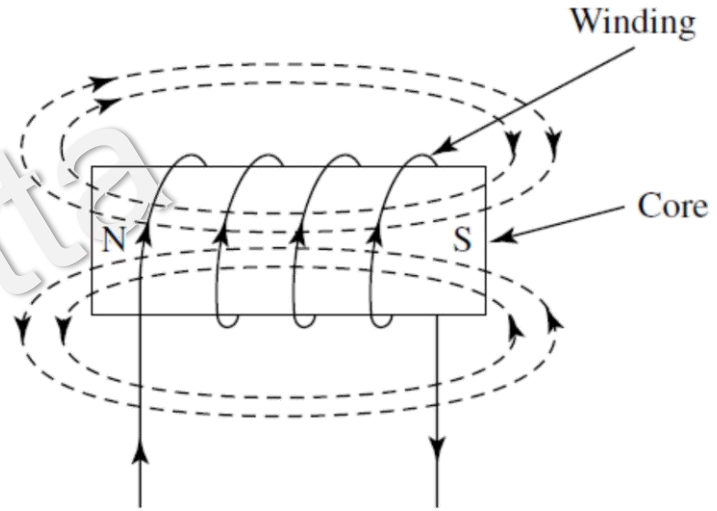
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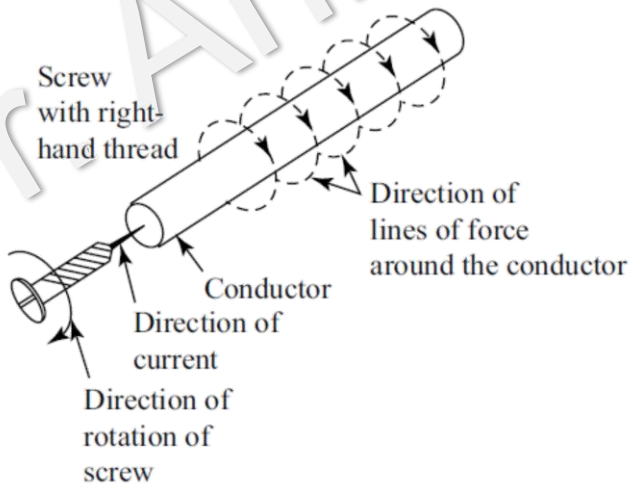
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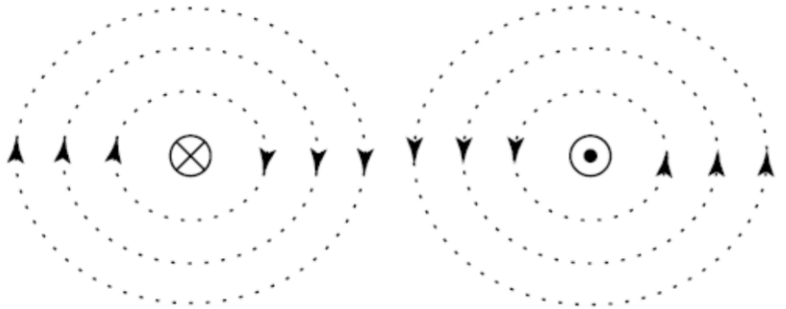
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Fundamentals of Electromagnetism

- Electromagnets are made by placing a coil around a magnetic material which forms the core. They demonstrate magnetic properties as long as current flows through the coil
- Magnetic field is the area around a magnet in which there is influence of the magnet, represented through lines of force or flux
- They come out of the magnet body from the North pole and enter the South pole and close their path through the magnet body
- The direction of current and flux around the coil sides are determined by applying the cork screw rule







current entering current leaving

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Fundamentals of Electromagnetism

- The strength of a magnetic field is expressed in terms of the number of **flux lines, φ** and is measured in Webers, where **1 Wb = 10^8 lines**

- **Magnetic flux density**, i.e., flux per unit area is denoted by B. If A is the area through which the flux lines emanate, i.e., come out, the flux density is given as

$$B = \frac{\phi}{A} \text{ Wb/m}^2 \text{ or Tesla}$$

- A **magnetic potential** or **magneto motive force (MMF)** is expressed as the product of the number of turns of the coil and the current passing through it, i.e.,

$$\text{MMF} = N\,I \text{ (Ampere Turns)}$$

- For the same MMF, the magnetic flux, φ or the magnetic flux density, B will be dependent upon the length of the field lines. The longer is the length, the weaker will be the flux density. **Magnetic field strength**, H is expressed as

$$H = \frac{NI}{l} \text{ AT/m}$$

- Since the magnetic field strength, H depends upon the number of turns of the magnetizing coil, the current flowing through the coil, and the flux line length, there would exist a relationship between the magnetic field strength H and the magnetic flux density B, when a magnetic material is magnetized by applying MMF.

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Fundamentals of Electromagnetism

- The flux density, B is related to the magnetizing force applied, H as:

$$B \propto H$$

$$B = \mu H$$

- Permeability (μ) is the ability of the material to allow the establishment of flux through it. It is similar to conductivity in an electrical circuit. It is the magnetic property of the core material

- The permeability of a magnetic material is compared with that of free space (μ_0) and is called relative permeability (μ_r). The relative permeability indicates how many times the material is more permeable than air.

$$\mu = \mu_o \mu_r$$

- For free space, μ_r of air or vacuum = 1. So:

$$\mu_0 = \frac{B}{H} \text{ henry per meter } = 4\pi \times 10^{-7} \text{ H/m}$$

- For other materials:

$$B = \mu_0 \mu_r H = \mu H$$

$$\mu = \mu_0 \mu_r = 4\pi \times 10^{-7} \times \mu_r$$

where

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Hysteresis Loss and Eddy Current Loss in Magnetic Materials

Hysteresis loss: When an alternating current flows through the coil, the core will get magnetized in the opposite directions alternately. The work done in this process will cause power loss, which is called hysteresis loss. The expression for hysteresis loss, W_h is given as

$$\text{Hysteresis loss, } W_h = K_h v f B_m^{1.6} \text{ W}$$

Where, K_h is a constant which depends upon the material and the range of flux density, v is the volume of the core material, f is the frequency of alternation of the current passing through the magnetizing coil, B_m is the maximum value of flux density in the core in Wb/m^2. Depending on the quality of the material, the power of B_m may vary from 1.5 to 2.0.

Eddy current loss: When a magnetic material is subjected to a changing magnetic field (e.g. the magnetic core of an inductor), EMF is induced in the core. This EMF causes circulating currents in the core. These circulating currents are called eddy currents. Energy in the form of heat is lost in the core due to this eddy current flow. Eddy current loss depends on the following:

(i) thickness of the magnetic material, t;

(ii) frequency of current producing the alternating magnetic field, i.e., the frequency of current flowing through the magnetizing coil, f;

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Hysteresis Loss and Eddy Current Loss in Magnetic Materials

(iii) maximum flux density, B_m

(iv) volume of the material, v

$$\text{Eddy current loss, } W_e = K_e v B_m^2 f^2 t^2 \text{ W}$$

Where K_e is the eddy current coefficient which depends upon the type of the magnetic material. To reduce eddy current loss in a magnetic material, the thickness is reduced. Laminated sheets are used to build a core instead of using one piece solid core.

The sum of hysteresis loss and eddy current loss is called core loss or iron loss.

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Self-induced EMF and Mutually-induced EMF

The EMF induced in a coil due to change in flux linkage when a changing current flows through the coil is called **self-induced EMF**.

When a second coil is brought near a coil producing changing flux, EMF will be induced in the second coil due to change in current in the first coil. This is called **mutually induced EMF**.

The magnitude of the induced EMF will depend upon the rate of change of flux linkage and the number of turns of the individual coils. The induced EMF in the two coils, e_1 and e_2 will be

$$e_1 = N_1 \frac{d\phi}{dt}$$

$$e_2 = N_2 \frac{d\phi}{dt}$$

where N_1 and N_2 are the number of turns of coil 1 and coil 2, respectively.

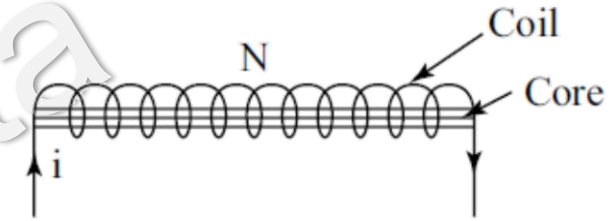
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Self-Induction of a Coil

Let us consider a coil of N turns wound on a core of magnetic material. Let an alternating current i pass through the coil

The emf induced, e will be

$$e = -N\frac{d\phi}{dt}$$



$$\text{The flux, } \phi = B \times A$$
$$= \mu HA$$
$$= \mu \frac{Ni}{\ell} A$$
$$= \frac{\mu NA}{\ell} i$$

where μ is the permeability of the core material; *l* is the length of flux path; A is the area of the coil. Substituting:

$$e = -N\frac{d}{dt}\left(\frac{\mu NA}{\ell} i\right)$$
$$= -\frac{\mu N^2 A}{\ell}\frac{di}{dt}$$
$$e = -L\frac{di}{dt}$$
$$L = \frac{\mu N^2 A}{\ell} \text{ Henry} \quad = \text{coefficient of self inductance or self-inductance}$$

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Self-Induction of a Coil

The inductance, L can be expressed in terms of the rate of change of the flux with respect to current flowing in the coil as:

$$\text{Flux, } \phi = BA$$
$$= \mu HA$$
$$= \mu \frac{Ni}{l} A$$

For a small increment of di, let the increase of flux be dφ. Therefore:

$$d\phi = \frac{\mu NA}{\ell} di$$
$$\frac{d\phi}{di} = \frac{\mu NA}{\ell}$$
$$N\frac{d\phi}{di} = \frac{\mu N^2 A}{\ell} = L$$
$$L = N\frac{d\phi}{di} \text{ Henry}$$

If φ and i have a linear relationship, $\frac{d\phi}{di}$ be written as $\frac{\phi}{I}$

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Self-Induction of a Coil

$$L = N\frac{\phi}{I} \text{ Henry}$$

$$L = \frac{\mu N^2 A}{\ell} = \frac{N^2}{\ell / \mu A} = \frac{N^2}{\text{Reluctance}}$$

$$\text{Reluctance} = \frac{\ell}{\mu A}$$ (Reluctance is the inverse of the permeability)

Again,

$$L = N\frac{\phi}{I}$$

$$LI = N\phi$$

Considering a small increase of i producing a small increase in φ as dφ, $Ldi = Nd\phi$

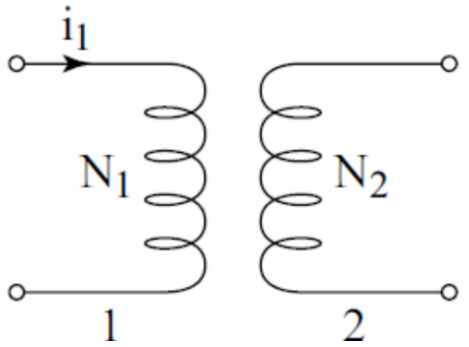
$$L\frac{di}{dt} = N\frac{d\phi}{dt} = e$$

Inductance is the property of a coil capable of inducing emf in itself due to changing current through it. A coil has an inductance of 1 Henry if a current of 1 Ampere flowing through the coil produces a flux linkage of 1 Wb-turn.

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Mutual Inductance

Let us consider two coils having N_1 and N_2 number of turns placed near each other. Let a changing current, i_1, flow through coil 1. The flux produced by i_1 in N_1 is φ_1. Since coil 2 is placed near coil 1, a part of the flux produced by coil 1 will be linked by coil 2. Let flux φ_2 linked by coil 2 is $\varphi_2 = K_1 \varphi_1$, where $K_1 \leq 1$. If magnetic coupling between the two is very good, the whole flux produced by coil 1 will link the coil 2, in which case the coefficient of the coupling K_1 will be 1. The induced EMF in coil 2 is e_2.



$$e_2 = N_2 \frac{d\phi_2}{dt} = N_2 \frac{d(K_1\phi_1)}{dt} = N_2 K_1 \frac{d\phi_1}{dt}$$

$$\phi_1 = \frac{\text{mmf}}{\text{Reluctance}} = \frac{N_1 i_1}{\ell_1 / \mu A_1} = \frac{N_1 \mu A_1}{\ell_1} i_1$$

$$e_2 = N_2 K_1 \frac{d}{dt}\left(\frac{N_1 \mu A_1 i_1}{\ell_1}\right) = \left(\frac{K_1 N_1 N_2 \mu A_1}{\ell_1}\right)\frac{di_1}{dt}$$

$$e_2 = M \frac{di_1}{dt}$$

$M = \dfrac{K_1 N_1 N_2 \mu A_1}{\ell_1}$, is called the mutual inductance of the two coils.

Similarly, if we calculate the induced EMF in coil 1, due to change in current in coil 2, we can find the induced EMF e_1 in coil 1 as

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Mutual Inductance

$$e_1 = \frac{K_2 N_1 N_2 \mu A_2}{\ell_2} \frac{di_2}{dt}$$

$$M = \frac{K_2 N_1 N_2 \mu A_2}{\ell_2}$$

$$M^2 = \frac{K_1 K_2 N_1 N_2 N_1 N_2 \mu A_1 \mu A_2}{\ell_1 \ell_2}$$

$$M^2 = K_1 K_2 \frac{\mu N_1^2 A_1}{\ell_1} \frac{\mu N_2^2 A_2}{\ell_2}$$

$$M^2 = K_1 K_2 L_1 L_2$$

$$M = \sqrt{K_1 K_2}\sqrt{L_1 L_2} = K\sqrt{L_1 L_2}$$

$$K = \sqrt{K_1 K_2}$$

$$M = K_1 N_2 \frac{N_1}{\ell_1 / \mu A_1}$$

$$\phi_1 = \frac{mmf}{Reluctance} = \frac{N_1 I_1}{\ell_1 / \mu A_1}$$

$$\frac{\phi_1}{I_1} = \frac{N_1}{\ell_1 / \mu A_1}$$

$$M = \frac{K_1 N_1 N_2 \mu A_1}{\ell_1}$$

$$= K_1 N_2 \frac{N_1}{(\ell_1/\mu A_1)}$$

$$M = K_1 N_2 \frac{\phi_1}{I_1}$$

$$M = N_2 \frac{\phi_2}{I_1} (\because \phi_2 = K_1 \phi_1)$$

$$M = \frac{N_2 \phi_2}{I_1} = \frac{\text{flux linkage in coil 2}}{\text{changing current in coil 1}}$$

Considering current change in the second coil,

$$M = N_1 \frac{K_2 \phi_2}{I_2}$$

$$M = N_1 \frac{\phi_1}{I_2} (\because \phi_1 = K_2 \phi_2)$$

$$M = \frac{N_1 \phi_1}{I_2} = \frac{\text{flux linkage in coil 1}}{\text{changing current in coil 2}}$$

So, the mutual inductance M between two coils is the flux linkage in one circuit due to change per unit of current in the other circuit.

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Problem

Q: A current of 5 A flowing through a coil of 500 turns produces a flux of 1 mWb. Another coil is placed near this coil and current in this coil is suddenly reversed in 10 ms. As a result, the EMF induced in the second coil is measured as 50 V. Calculate self and mutual inductance of the coils assuming a coefficient of coupling as 60 %.

$$e_2 = M\frac{di_1}{dt}$$

$$50 = M\frac{5+5}{10\times10^{-3}} \qquad [+5\text{ A current has been changed to } -5\text{ A}]$$

$$M = \frac{50\times10\times10^{-3}}{10}$$

$$M = 50\times10^{-3}\text{ H}$$

Self-inductance of coil 1 is

$$L_1 = \frac{N_1\phi_1}{I_1} = \frac{500\times1\times10^{-3}}{5} = 10\times10^{-3}\text{ H}$$

Using the formula,

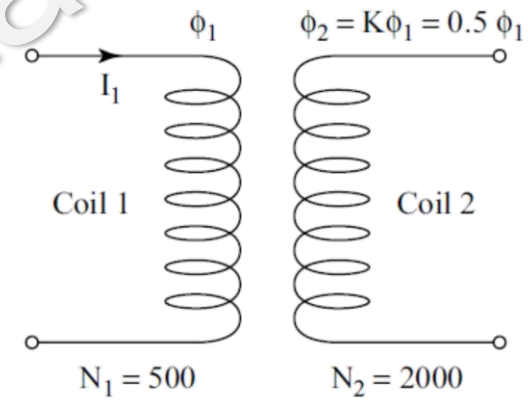
$$M = K\sqrt{L_1L_2}$$

$$50\times10^{-3} = 0.6\sqrt{10\times10^{-3}\times L_2} \qquad L_2 = 694.4\times10^{-3}\text{ H}$$

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Problem

Q: There is mutual magnetic coupling between two coils of number of turns 500 and 2000, respectively. Only 50% of the flux produced by the coil of 500 turns is linked with the coil of 2000 turns. Calculate the mutual inductance of the two coils. Also calculate the EMF induced in the coil of 2000 turns when current changes at the rate of 10A/s in the other coil. The self-inductance of the coil of 500 turns is 200 mH.



$$M = \frac{\text{flux linkage in coil 2}}{\text{current change in coil 1}}$$

$$= N_2 \frac{K\phi_1}{I_1} = 2000 \times \frac{0.5 \times \phi_1}{I_1} = \frac{1000\phi_1}{I_1}$$

$$L_1 = \frac{N_1 \phi_1}{I_1}$$

$$\frac{\phi_1}{I_1} = \frac{L_1}{N_1} = \frac{200 \times 10^{-3}}{500}$$

$$M = 1000 \times \frac{\phi_1}{N_1} = \frac{1000 \times 200 \times 10^{-3}}{500}$$

$$M = 400 \times 10^{-3}\ \text{H}$$

$$e_2 = M \frac{di}{dt}$$

$$= 400 \times 10^{-3} \times 10$$

$$= 4\ \text{V}$$

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ELECTRIC CIRCUITS AND SYSTEMS

Module 3 – Lecture 2

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Magnetic Circuits

- All electrical machines and equipment are made of magnetic material as their core. A winding which carries current is placed around the core. The core and the current-carrying coil around the core forms an electro magnet.
- The strength of this electro magnet produced by the magnetizing force is directly proportional to the magnetizing force and inversely proportional to the reluctance of the flux path.
- Reluctance is the opposition offered to the establishment of flux.
- The amount of flux produced by the magnet indicates the strength of the magnet.
- The more the magnetizing force (MMF), more is the flux produced.
- The more the opposition to flux path (i.e., reluctance or magnetic resistance) less is the flux produced.

$$\text{Flux} = \frac{\text{MMF}}{\text{Reluctance}}$$

$$\phi = \frac{NI}{S}$$

Reluctance $S = K\frac{l}{A} = \frac{l}{\mu A}$

$$K = \frac{1}{\mu}$$

l = length of the flux path
A = area of cross section of the flux path
μ = permeability of the magnetic material.

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Magnetic Circuits

For the same amount of ampere turns, the flux produced by an iron-core coil is much more than that produced by an air-core one. If we make a cut in the magnetic material to create an air gap, the flux produced for the same ampere turns will be somewhat less because the total reluctance of the flux path is now increased. The flux has to cross the air gap whose reluctance is very high as compared to iron. The flux produced will be calculated as

$$\text{Flux, } \phi = \frac{\text{MMF}}{\text{Reluctance of iron path} + \text{Reluctance of air gap}}$$

To calculate the ampere turns required to create a required amount of flux we use the relations:

$$H = \frac{NI}{l} = \frac{AT}{l} \text{ or } AT = H \times l$$

$$\phi = \frac{\text{MMF}}{s} = \frac{AT}{l/\mu A} = \frac{AT}{l}\mu A = H\mu A$$

$$\frac{\phi}{A} = \mu H$$

$$B = \mu H \text{ and } AT = H \times l$$

The various quantities associated with magnetic circuits are stated as follows.

Magneto motive force (MMF) = Ampere turns = N I

$$\text{Magnetic field strength } (H) = AT / m = \frac{NI}{l}$$

$$\text{Flux density (B) = Flux per unit area} = \frac{\phi}{A}$$

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Magnetic Circuits

Permeability $\mu = \mu_o \mu_r$

Permeability of free space (air), $\mu_o = 4\pi \times 10^{-7}$

Relative permeability (μ_r) = The number of times the material is more permeable than air.

Flux density, $B = \mu H$

Flux, $$\phi = BA = \mu HA = \mu_o \mu_r \frac{NI}{l} A$$

$$= \frac{NI}{l / \mu_o \mu_r A}$$

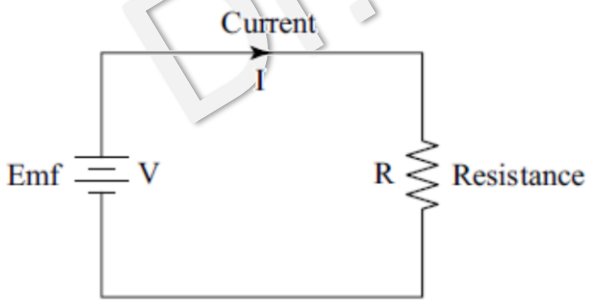
$$= \frac{MMF}{S}$$

Reluctance, $$S = \frac{l}{\mu_o \mu_r A}$$

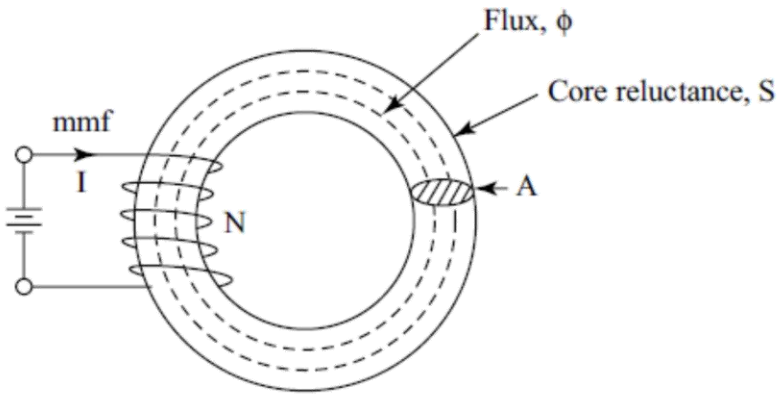
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Analogy between Magnetic and Electric circuits

Electric Circuit	Magnetic Circuit
Current, I	Flux, ϕ
EMF, V	MMF, N I
Resistance, R	Reluctance, S
$I = \frac{EMF}{Resistance} = \frac{V}{R}$	$\phi = \frac{MMF}{Reluctance} = \frac{NI}{S}$
$R = \rho\frac{l}{a}$	$S = \frac{l}{\mu A}$
$Conductance = \frac{1}{Resistance}$	$Permeance = \frac{1}{Reluctance}$
Electric field intensity $= \frac{V}{l}$	Magnetic field intensity, $H = \frac{AT}{l}$ or, $H = \frac{NI}{l}$



Electric Circuit



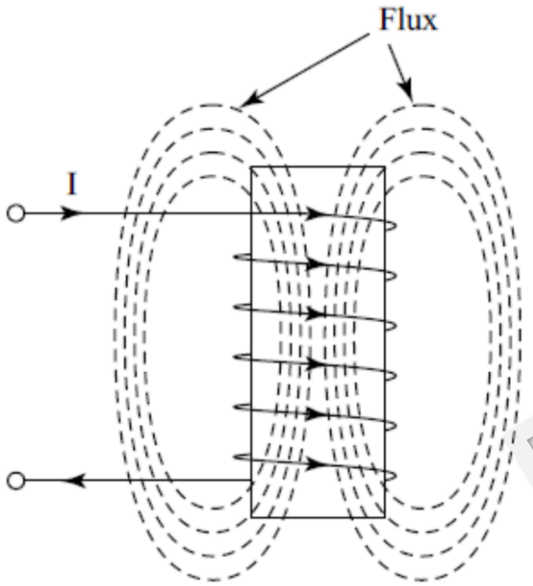
Magnetic Circuit

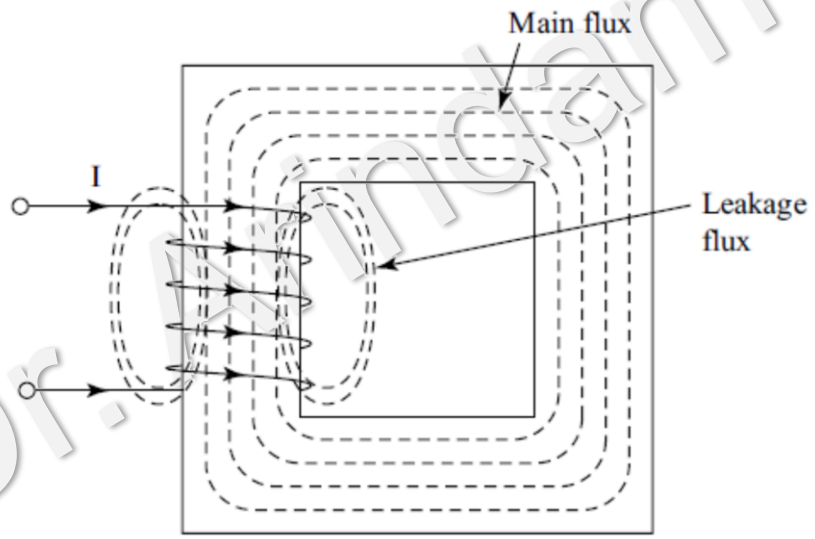
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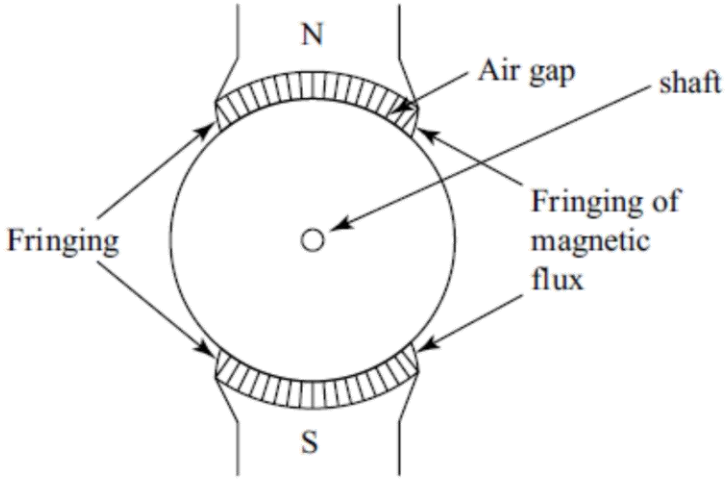
Magnetic Leakage and Fringing

Most of the flux will flow or pass through the magnetic material which is called the main flux or useful flux. However, a certain percentage of flux will link the coil itself and will not pass through the entire core. This flux is called **leakage flux.** Leakage flux completes its path through air instead of going through the entire iron path.

In a magnetic circuit with some air gap, when flux has to pass through the air gap, there is tendency of the magnetic flux to spread out at the two edges or sides. This effect is called **fringing**.





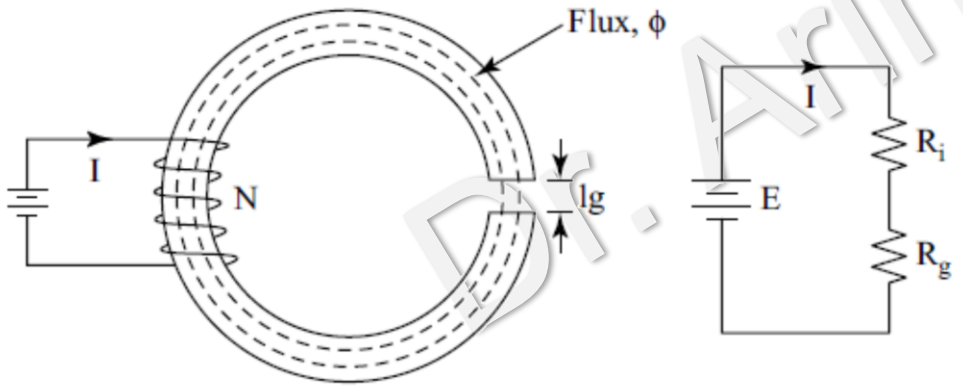


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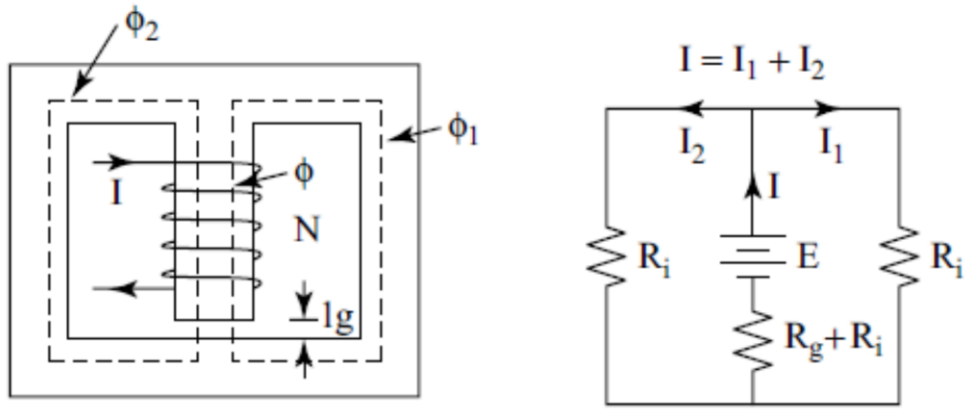
Series and Parallel Magnetic Circuits

The equivalence of MMF (NI), of the magnetic circuit is EMF (E or V) of the electric circuit. Magnetic reluctance of the iron path and that of air gap are represented by their equivalent resistances R_i and R_g in the electric circuit. Flux in the magnetic circuit is represented by current in the electric circuit. The magnetizing force, NI / *l* produces the flux in the core. The opposition to the flux path is provided by iron and the air in the air gap through which the flux has to pass. The reluctance of iron is very low whereas that of air is very high. For a small amount of air gap, the reluctance will be much more than along the length of the iron path.

Due to spreading of the flux, the effective area of cross section of the air gap through which the flux passes increases, which is not desirable.



(a) Series circuit



(b) Parallel circuit

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Problems

Q: A circular iron ring of mean diameter 25 cm and cross-sectional area 9 cm^2 is wound with a coil of 100 turns and carries a current of 1.5 A. The relative permeability of iron is 2000. Calculate the amount of flux produced in the ring.

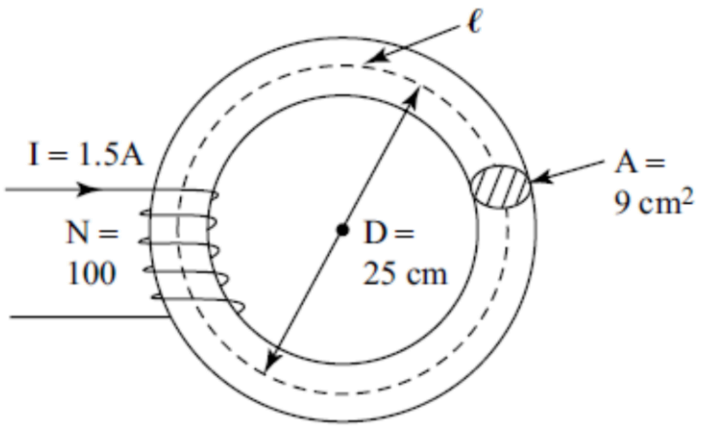
Mean length of flux path, $\ell = \pi D = 3.14 \times 25 \text{ cm} = \frac{3.14 \times 25}{100} \text{ m} = 0.785 \text{ m}$

Flux, $$\phi = \frac{\text{MMF}}{\text{Reluctance}} = \frac{\text{NI}}{\ell/\mu_o \mu_r A} = \frac{100 \times 1.5}{0.785/4\pi \times 10^{-7} \times 2000 \times 9 \times 10^{-4}}$$

$$= \frac{150}{348 \times 10^3} = \frac{150}{348} \times 10^{-3}$$

$$= 0.431 \times 10^{-3} \text{ Wb}$$

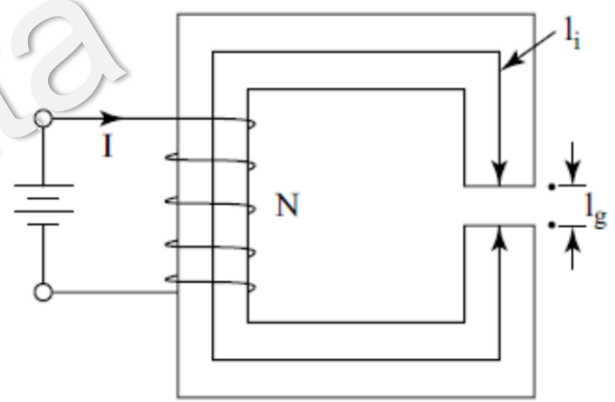
$$= 0.431 \text{ mWb}$$



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Problems

Q: A rectangular shape iron core has an air gap of 0.01 cm. The mean length of the flux path through iron is 39.09 cm. The relative permeability of iron is 2000. The coil has 1000 turns. The cross-sectional area of the core is 9 cm^2. Calculate the current required to produce a flux of 1 mWb in the core.



Total reluctance of the flux path = Reluctance of iron path + Reluctance of air gap,

i.e., $$S = S_i + S_g$$

$$S = \frac{l_i}{\mu_o \mu_r A} + \frac{l_g}{\mu_o A}$$

Note: the iron path permeability is μ which is equal to $\mu_0\mu_r$ whereas for the air gap the permeability is μ_0 only.

$$S = \frac{39.09 \times 10^{-2}}{4\pi \times 10^{-7} \times 2000 \times 9 \times 10^{-4}} + \frac{0.01 \times 10^{-2}}{4\pi \times 10^{-7} \times 9 \times 10^{-4}}$$

$$= \frac{10^6}{4\pi}\left[\frac{39.09}{18} + \frac{100}{10}\right] = \frac{295.45 \times 10^5}{36\pi}$$

$$\phi = \frac{NI}{S} = \frac{1000\ I}{S}$$

$$I = \frac{\phi \times S}{1000} = \frac{1 \times 10^{-3} \times 295.45 \times 10^5}{36\pi \times 1000} = \frac{29.545}{36\pi} = 0.26\,A$$

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Problems

Q: Neglecting fringing effects, calculate the flux produced in the air gap in the magnetic circuit, which is excited by the MMF of two windings. The mean length of the flux path is 40 cm. The permeability of iron is 2000. The uniform core cross-sectional area is 10 cm^2.

Flux produced by the MMF of the two coils are in opposite directions. Thus, resultant MMF is the difference of the two MMFs.

Total MMF $= N_1 I_1 - N_2 I_2$

$= 100 \times 10 - 80 \times 1.5$

$= 880$ AT

Total reluctance = Reluctance of iron + Reluctance of air gap

$$= \frac{l_i}{\mu_o \mu_r A} + \frac{l_g}{\mu_o A} = \frac{1}{\mu_o A}\left[\frac{l_i}{\mu_r} + l_g\right]$$

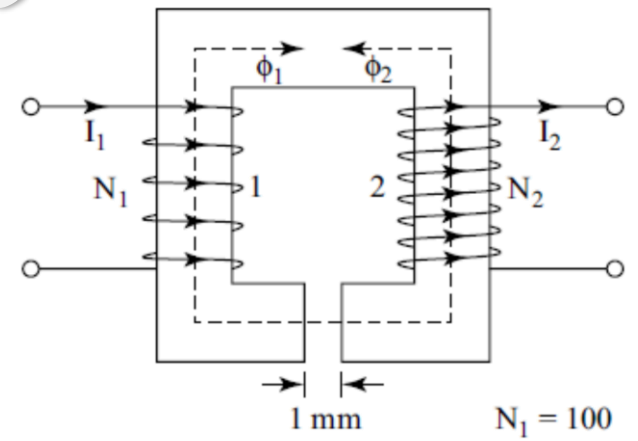
Substituting values

Total Reluctance $$= \frac{1}{4\pi \times 10^{-7} \times 10 \times 10^{-4}}\left[\frac{(40 \times 10^{-2} - 1 \times 10^{-3})}{2000} + 1 \times 10^{-3}\right]$$

$$= \frac{10^{10}}{4\pi}\left[\frac{399 \times 10^{-3}}{2000} + 1 \times 10^{-3}\right]$$

$$= \frac{10^7}{4\pi}\left[\frac{399}{2000} + 1\right]$$

$= 0.955 \times 10^6$ AT/Wb.



$N_1 = 100$
$I_1 = 10$ A
$N_2 = 80$
$I_2 = 1.5$ A

Core Flux $$= \frac{\text{MMF}}{\text{Reluctance}} = \frac{880}{0.955 \times 10^6} = 921.5 \times 10^{-6}$$

$= 0.9215 \times 10^{-3}$ Wb

= air-gap flux (since no fringing)

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EEE1001
ELECTRIC CIRCUITS AND SYSTEMS

Module 3 – Lecture 3

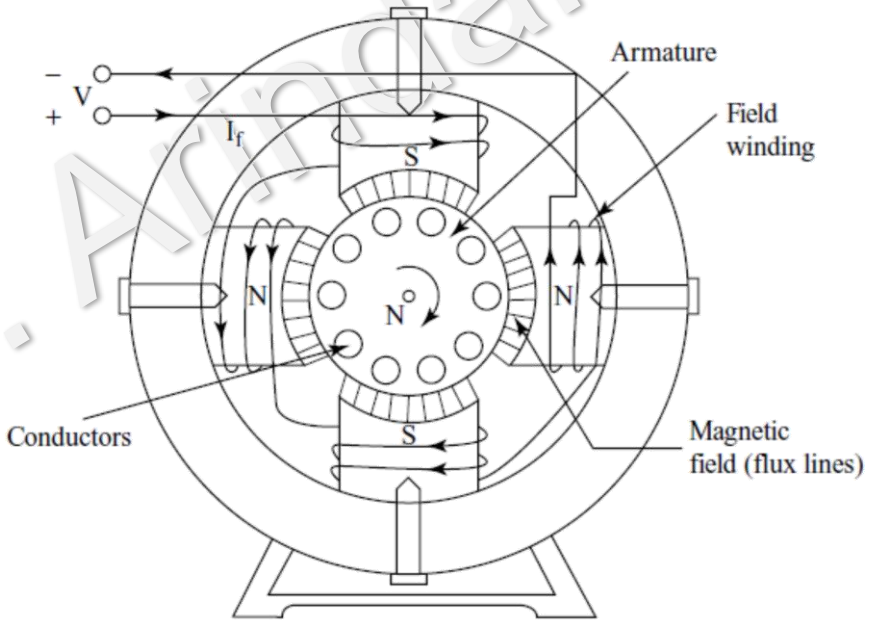
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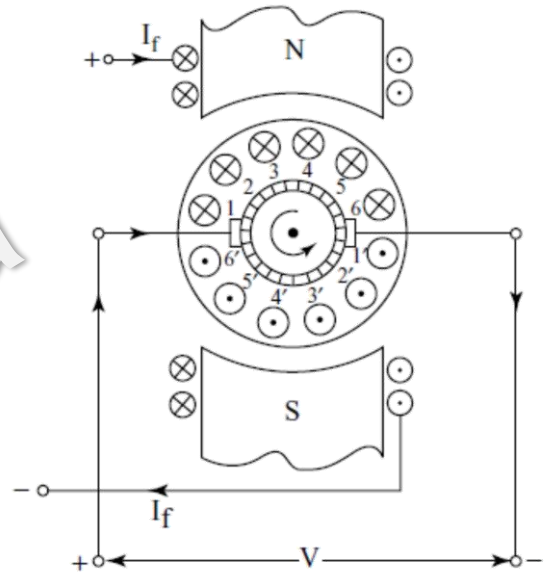
Working Principle of DC Generators

DC machines work either as a dc generator or as a dc motor. In a dc generator, a set of conductors or coils placed on a rotating body, called armature, are rotated continuously inside a magnetic field with the help of a prime mover (prime mover is another machine, may be a diesel engine or a turbine, which rotates the armature). The magnetic field is created by passing dc current through the windings of a set of field magnets. When conductors pass under alternate North and South poles, alternating EMF is induced in the armature winding. The ac generated gets converted into dc when the voltage is collected from the rotating armature through the brush and commutator arrangement. The brush and commutator arrangement, therefore, works like a full-wave rectifier which converts generated ac into dc for the output circuit. A dc generator, therefore, converts mechanical energy supplied through the prime mover to electrical energy to be supplied from the generator armature to an electrical load.



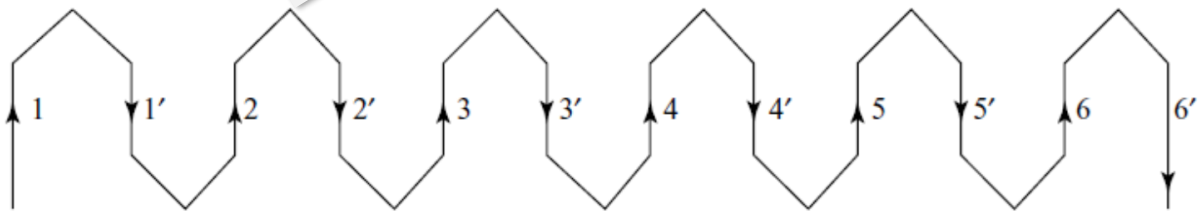
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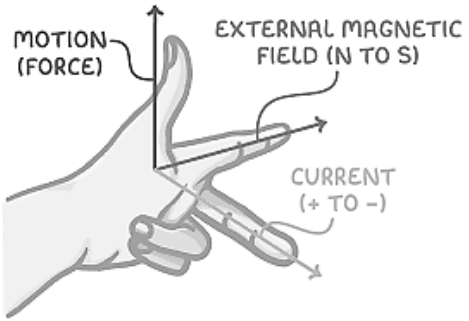
Working Principle of DC Motors



A dc motor works on the basic principle that when a current-carrying conductor is placed in a magnetic field it experiences a force. In a dc motor, the armature carries a number of conductors placed in slots and the armature is placed inside the magnetic field created by field magnets. The field magnets are excited by field current passing through them. The field windings may either be excited from a separate dc source or could be excited from the source of supply provided to the armature. The armature is fed from a dc supply source of V Volts as shown. The direction of current flowing through the winding have been shown by cross ⊗ and dot ⊙. A cross indicates current flowing in the direction perpendicular to the plane of the paper downwards and a dot indicates that current is coming towards the observer looking into the paper.

There are six coils 1−1′, 2−2′, 3−3′, 4−4′, 5−5′, and 6−6′ placed in armature slots. Current in coil sides 1, 2, 3, etc. are shown by cross and current at the other coil sides, i.e., 1′, 2′, 3′, etc. are shown by dots. By applying Fleming's Left-Hand rule, it is seen that force developed on the upper half of the armature conductors is from right to left and for the lower half of the armature conductors it is from left to right. These forces lead to development of torque which causes rotation of the armature in the anticlockwise direction. It may be noted carefully from the figure that as the armature rotates in the anticlockwise direction, current in the conductors of a coil passing under the brushes would change. This will ensure that at any point of time any conductor on the upper half of the armature will have cross current and on the lower half will have dot current. There will be continuous rotation of the armature in one direction. The reversal of current in the coils passing under the brush is called commutation.



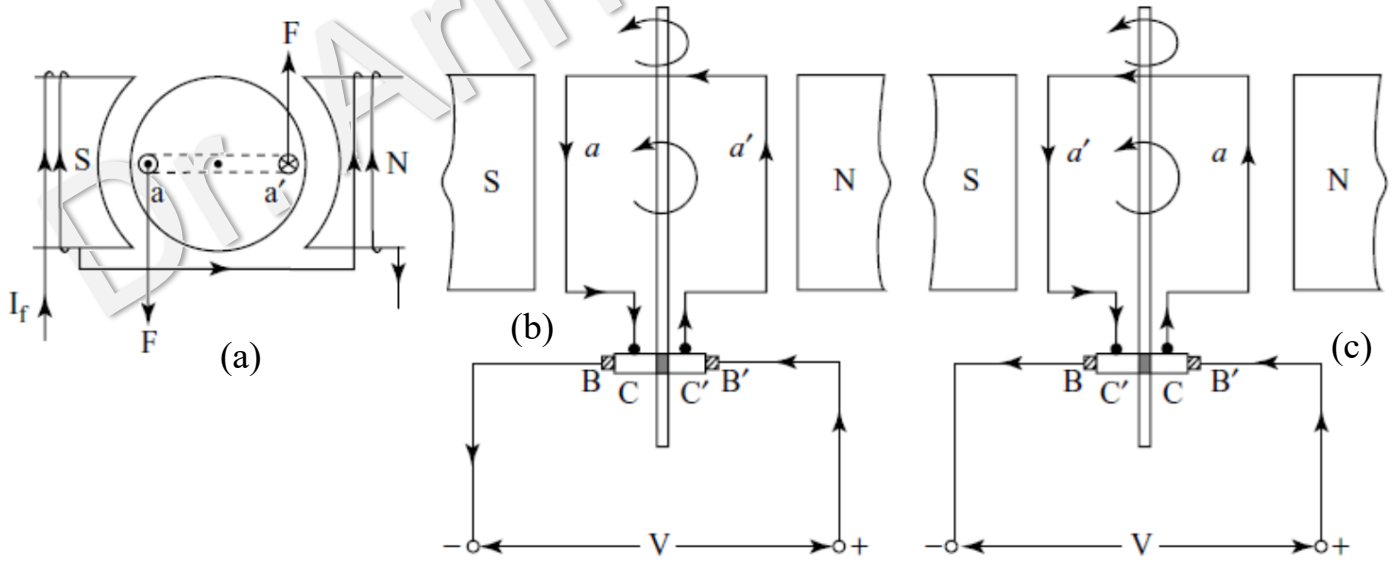


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Function of Brush and Commutators

The armature having the coil on it is placed in a magnetic field created by the field system as shown in Fig. (a). In Fig. (b) is shown how current will flow from the positive polarity of the supply source through brush B′ and commutator segment C′ to coil side a′ and then to coil side a, returning through commutator segment C and brush B to the negative terminal of the supply source. By applying Fleming's Left Hand rule to coil-side a′ in Fig. (b) we find that the conductor will experience an upward force whereas the coil side a will experience a downward force. These two forces would create a torque to rotate the armature in the anticlockwise direction. After every half revolution, i.e., for every rotation of 180° mechanical, coil side a′ along with the connected commutator segment C′ will change positions with the coil side a and the connected commutator segment C. After half revolution it is seen that the direction of current in the coil has reversed. Earlier as in Fig. (b), current was flowing from a′ to a and after half revolution current is flowing from a to a′ as shown in Fig. (c). Supply polarities remaining fixed, it is seen that current in the armature coil is alternating its direction. However, the direction of the rotation of the coil, as obtained, is unidirectional, i.e., in this case in the anti clockwise direction. The torque developed will be alternating in every half cycle thereby net torque being zero. Thus, the function of brush and commutator in a dc motor is to produce unidirectional torque, i.e., to cause rotation of the armature of the motor in a particular direction.

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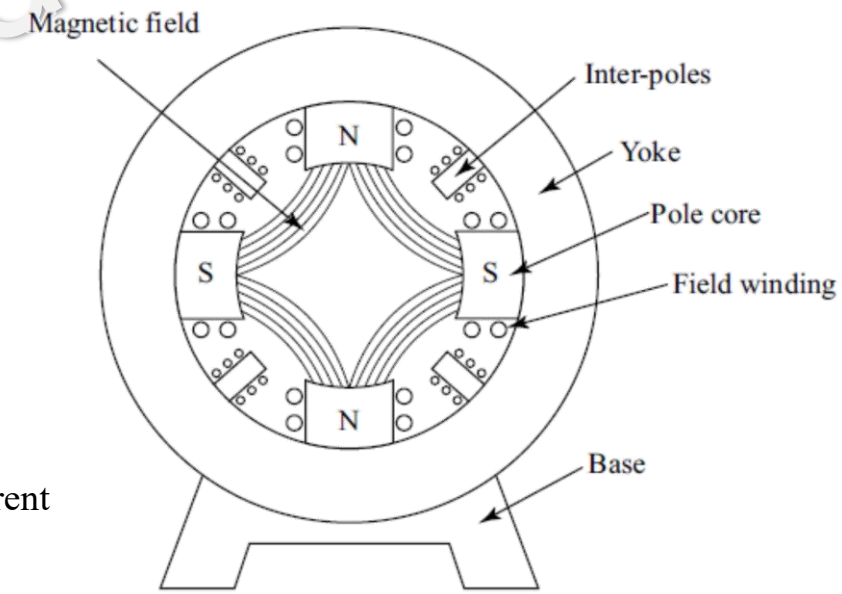


Constructional Details of DC Machine

A dc machine consists of a field system which produces the magnetic field, the armature which carries the armature conductors placed in slots, the brush and commutator arrangement, the shaft, and the bearings.

Field System: The purpose of the field system is to produce a magnetic field inside which a set of conductors will be rotating. The field system consist of a set of electromagnets fixed on the inside periphery of a hollow cylindrical structure called the yoke. The field poles have field windings wound on a laminated iron core. The number of poles of a dc machine may be two or multiples of two. A dc current supplied from a dc source magnetizes the field system. Alternate North and South poles are formed on the basis of the direction of the current flowing through the field windings. Small poles, called interpoles are often fixed between two main poles, particularly in case of large dc machines. The field windings are made of thin insulated copper wire of a large number of turns. The resistance of field winding is fairly high of the order of 100 Ω or so.

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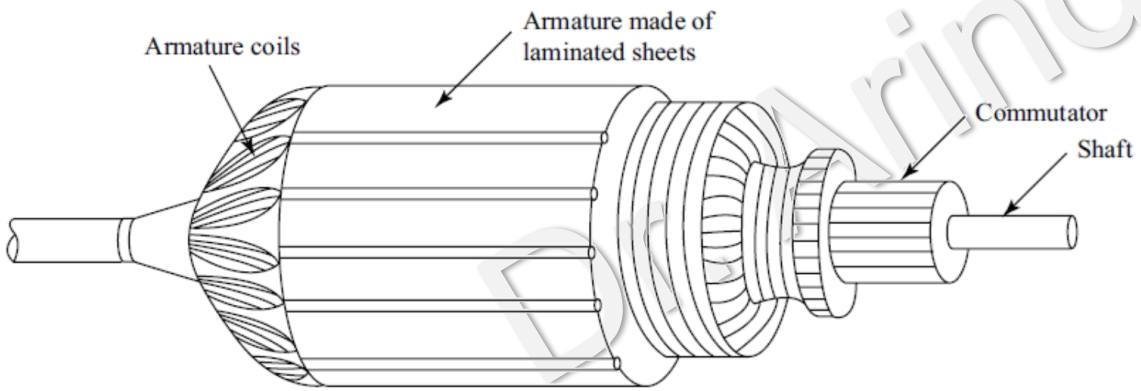
The field system creating a magnetic field when current will flow through the field windings

Constructional Details of DC Machine

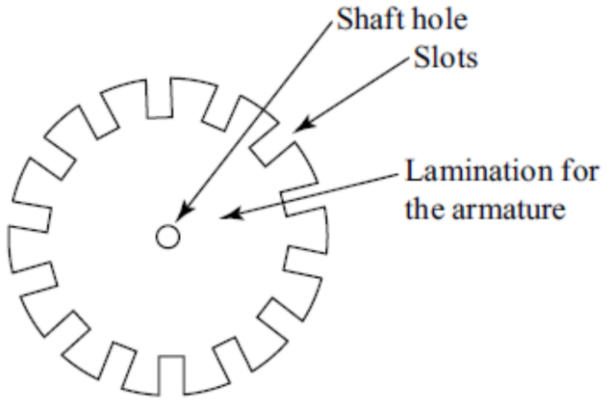
Armature: The armature of a dc machine is built by using circular laminated sheet steel to form a cylindrical structure with a shaft passing through its centre.

In a dc machine the field system is stationary while the armature along with the commutator is the rotating part. The armature winding is made using a large number of coils connected in series and parallel to get the desired voltage and current. The coils are made of insulated copper wires and are placed in a large number of armature slots. The coil ends are connected to large number of commutator segments of the commutator. The commutator segments are insulated from each other using some good quality thin insulating sheets like mica sheets. Carbon brushes are placed on the commutator surface and terminals are brought out from the brushes.

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Armature made of laminated sheets and the commutator assembly

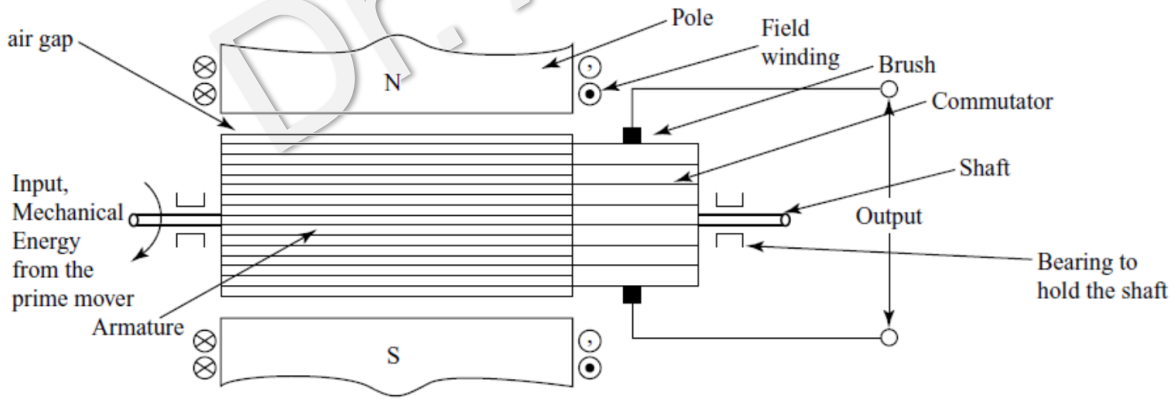


Laminated sheets used to make the cylindrical armature

Constructional Details of DC Machine

The air-gap between the field poles and the armature is kept small, of the order of few mm. The commutator, like the armature, is cylindrical in shape and is made up of a large number of wedge-shaped segments of hard drawn copper. The fixed carbon brushes sitting on the commutator surface make slipping contacts with the armature coils via the commutator segments either to collect current or to supply current from an external source (in case of motoring operation). Brushes made of carbon are conducting material but softer than the hard drawn copper used in the commutator. Brushes are placed in brush holders which are fixed with the stationary part of the machine. When the armature rotates, the brushes and the commutator surface make constant smooth rubbing, which over a period of time reduces the length of the brushes (due to wear and tear). There is no deterioration of the surface of the commutator due to this rubbing action. When required, the set of brushes can be replaced by new ones.

Armature and the commutator together is made into one unit. A shaft made of mild steel runs through the armature and comes out from both sides. Two sets of bearings are used to support the whole of the revolving system. The shaft extension on one end is used to connect the prime mover while at the commutator end the shaft is extended for use of the bearing. The shaft is held in position inside the stator with the help of end shields.

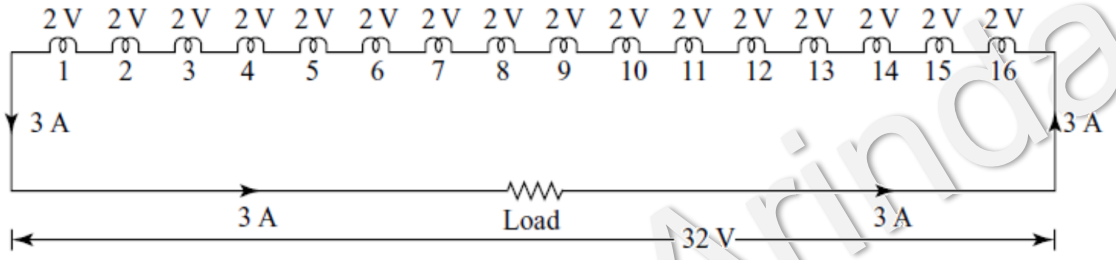


DC generator

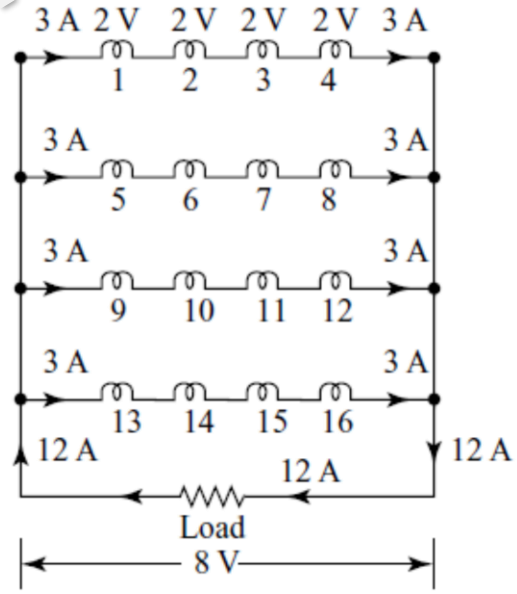
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Constructional Details of DC Machine

Armature Winding: When the armature coils rotate in the magnetic field, EMF is induced in each coil. If all the coils are connected in series, the total EMF available will be the sum of all the EMFs in all the coils. But the current that this winding would be able to deliver to the load will be governed by the current-carrying capacity of each of the armature coils. If higher current is desired, the coils have to be connected in series and parallel. Thus, the armature winding will have a set of coils connected in series in each of its parallel paths. The arrangement is exactly similar to series–parallel connection of cells used to make a battery of certain voltage and ampere rating.



Series connection of all the armature coils



Series–parallel connection of the coils

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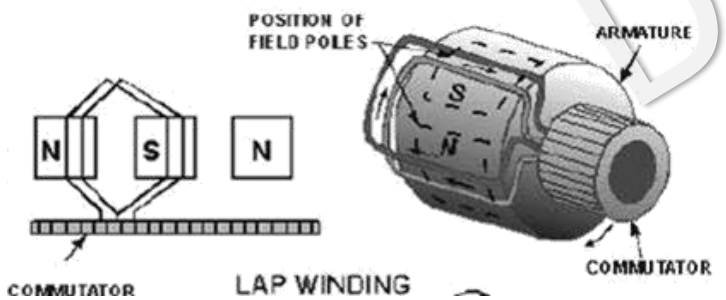
Constructional Details of DC Machine

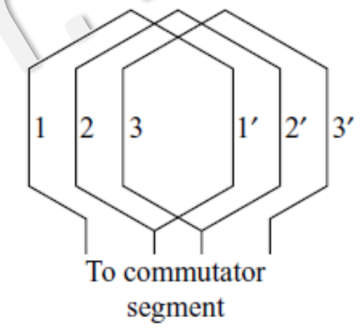
Types of Armature Winding: All the coils placed in armature slots are connected together in a particular manner to form the armature winding. Two basic types of winding connections are made. They are (i) lap winding and (ii) wave winding. In all cases, the coil ends are connected to commutator segments. The commutator segments are, in fact, extension of coil end connections. Brushes placed on commutator touching the commutator segments make slipping contact with the coils. The two types of armature windings are explained below.

(a) Lap winding: Here, the end of one coil is connected to the beginning of the next coil. The number of parallel paths formed is equal to the number of poles as the winding divides itself naturally into several parallel circuits, one for each pole.

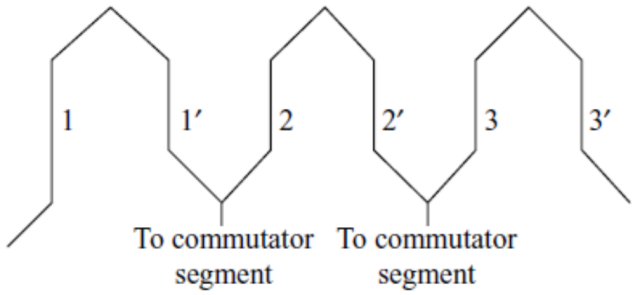
(b) Wave winding: The end of one coil is connected to the next coil side under the next similar poles. The winding so formed looks like a wave, and hence the name. The number of parallel paths formed is always equal to 2 as it connects all the coils with a clockwise current into one series path and all the coils with an anti-clockwise current into a second series path. Note that the coils are connected to commutator segments in both types of windings.

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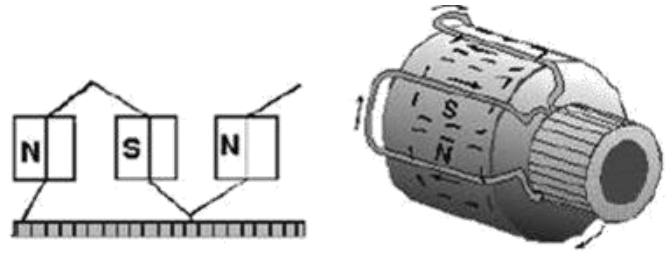




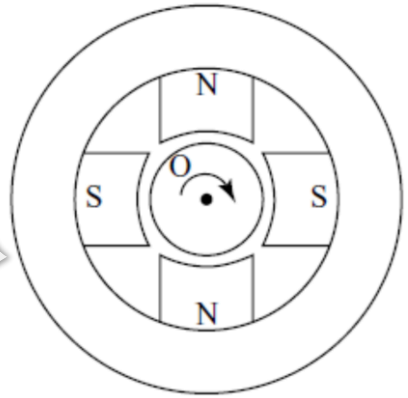
(a) Lap



(b) Wave



EMF Equation of DC Machine



No. of poles = P
Flux per pole = φ Wb
Speed of the driving prime mover = N rpm or N/60 rps
Time taken by the conductor to make 1 revolution = 60/N s
Actual number of armature conductors = Z
Number of parallel paths of the armature winding = A
Induced EMF in the conductors will be due to relative velocity between the conductor and the flux produced by the field poles
When the conductor makes 1 revolution, the flux cut by the conductor = Pφ Wb

Therefore, induced EMF in 1 conductor

$$= \frac{\text{Flux cut in 1 revolution in Wbs}}{\text{Time taken in making 1 revolution in secs}}$$

$$= \frac{P\phi}{60/N} = \frac{P\phi N}{60} \text{ V}$$

The induced EMF available across the output terminals will be equal to the induced EMF per parallel path.

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EMF Equation of DC Machine

Thus, total induced EMF, E:

$$E = \frac{P\phi N}{60}\left(\frac{Z}{A}\right) V$$

$$E = \frac{\phi ZNP}{60A} V$$

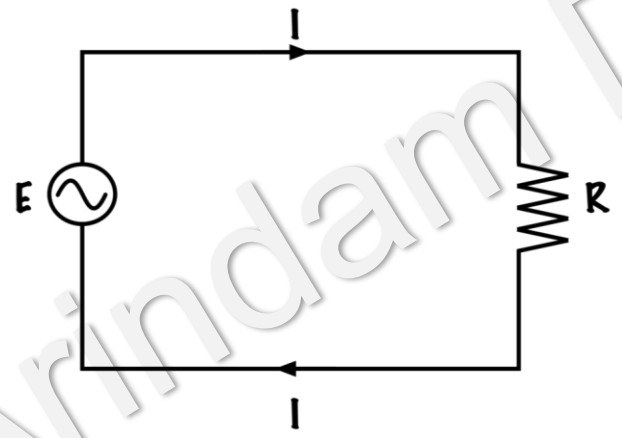
When the dc machine is also working as a motor, the current-carrying conductors placed in the magnetic field will develop torque and rotate in a particular direction. When they would rotate, EMF will also be induced in them.

The equation for induced EMF will be the same both in generating and motoring mode of operation of the machine. The induced EMF in the armature of a dc motor is often called back EMF as it opposes the applied voltage.

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Single Phase System

In electricity, phase refers to the distribution of a load. Single-phase system is a two-wire ac power circuit. Typically, there is one power wire—the phase wire—and one neutral wire, with current flowing between the power wire (through the load) and the neutral wire. In other words, one wire is for the power to go in and the other is a return path for current. Basically, current only has one path to travel in a single-phase circuit. The circuits we have dealt with so far are single-phase circuits.



Single-phase power supplies are most commonly used when typical loads are lighting or heating, rather than large electric motors. Therefore, residential homes are usually served by a single-phase power supply.

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Three Phase System

Generation, transmission, and distribution of electricity is done by three-phase electrical networks consisting of generators, transformers, and transmission and distribution lines forming the power system. In a three-phase system we have three independent voltages induced in the three windings of the generator. In fact, a three-phase system has four wires, i.e., the three current carrying conductors and one neutral for passing the fault current to the earth. The cross-sectional area of the neutral conductor is half of the live wire. The current in the neutral wire is equal to the sum of the line current of the three wires.

Advantages over single phase:

(i) the output electricity is more than the output of a single-phase machine of the same size;

(ii) the most commonly used three-phase induction motors are self starting. For single-phase motors a separate starting winding is required;

(iii) power transmission is more economical than single phase power transmission;

(iv) the power factor is better than that of the single-phase systems;

(v) single-phase supply can also be obtained from a three-phase supply;

(vi) the instantaneous power in a single-phase system is fluctuating with time giving rise to noisy performance of single-phase motors. The power output of a symmetrical three-phase system is steady;

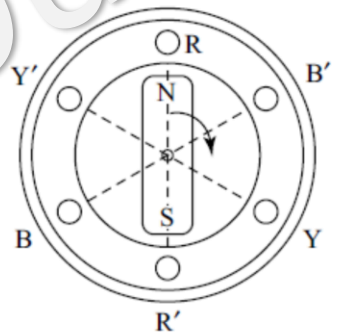
(vii) for rectification of ac into dc, the dc output voltage becomes less fluctuating if the number of phases is increased.

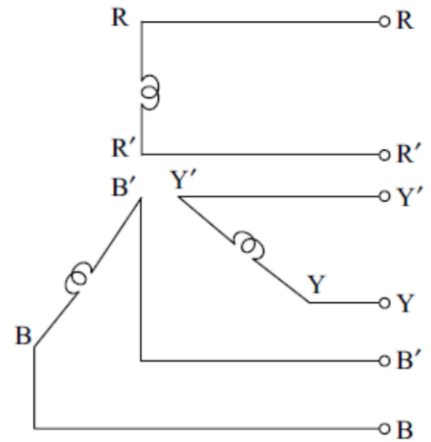
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Three Phase System

Commercial generators are built to generate three-phase voltages. In three-phase generators, three separate windings are made. Windings are made of coils. These windings are placed in stator slots at an angle of 120˚ apart. Thus, across the three-phase windings we will get three voltages which are equal in magnitude and frequency but having a time phase difference of 120° between them.

$$e_R = E_m \sin \omega t$$
$$e_Y = E_m \sin (\omega t - 120°)$$
$$e_B = E_m \sin (\omega t - 240°)$$

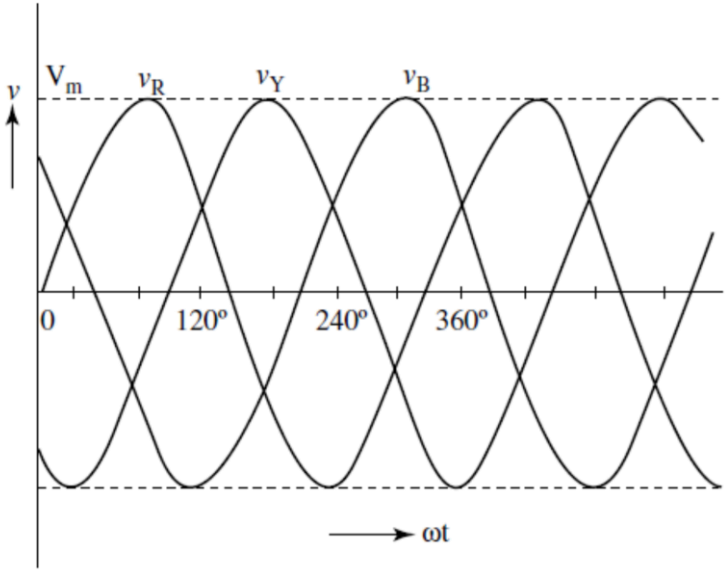


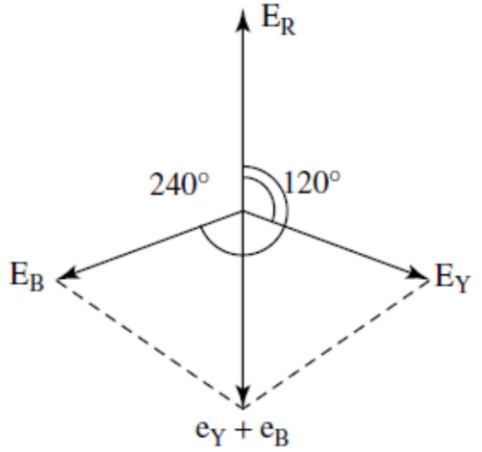


Since the three voltages are equal in magnitude but displaced in time phase by 120˚, their phasor sum is zero.

$$\text{Resultant EMF} = E_m \sin \omega t + E_m \sin (\omega t - 120°) + E_m \sin (\omega t - 240°) = 0$$

The order in which the phase voltages attain their maximum value or peak value, V_m is called the phase sequence. If the rotor rotates in the clockwise direction, voltages in the phases will be induced in the sequence, V_R, V_Y, V_B. If the rotor poles rotate in the opposite direction the phase sequence of the induced voltage will change from RYB to RBY.





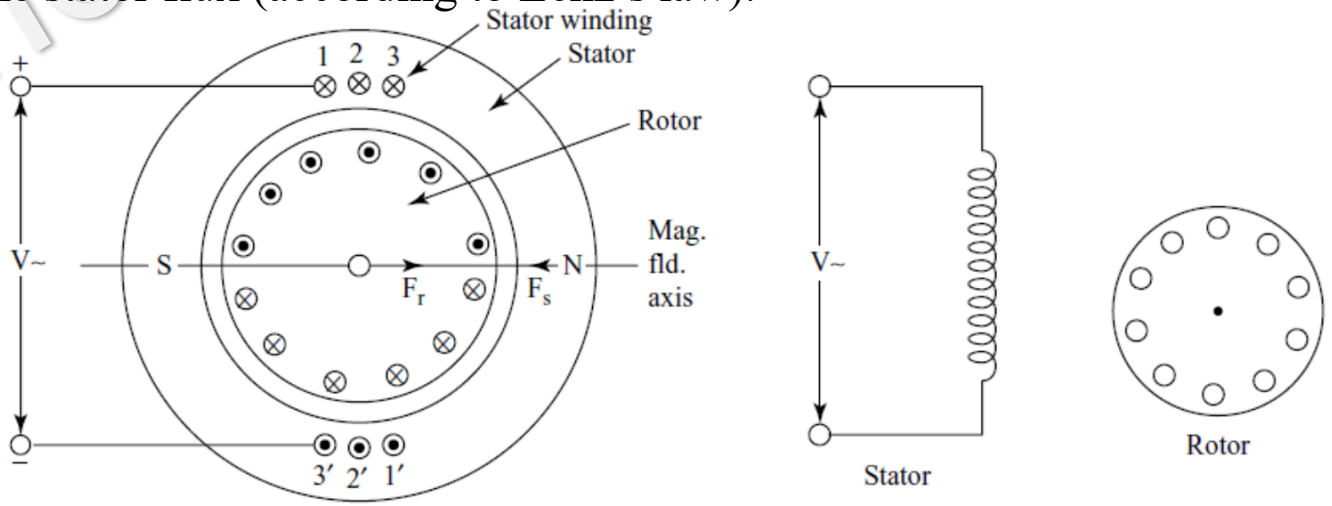
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Single Phase Induction Motor

Single-phase induction motors are widely used in electrical appliances and gadgets like ceiling fans, exhaust fans, refrigerators, washing machines, etc. They require only single-phase supply to the stator. The rotor is squirrel-cage type and does not require any supply from a separate source.

Constructional details and working principle: The stator of a single-phase induction motor has a single-phase winding and the rotor is of squirrel-cage type. These motors do not develop starting torque unless some mechanism for starting the motor is provided. An auxiliary winding is provided in the stator for developing starting torque. When supply from a single-phase source is applied, current will flow through the stator winding for the instantaneous polarity of voltage. Since the supply voltage is varying sinusoidally, the magnitude and direction of the flux produced will change. EMF will be induced in the rotor winding. The rotor winding being a closed winding, current will flow through it. The direction of flux produced will be such that the rotor flux will oppose the stator flux (according to Lenz's law).

Thus, the two magnetic fields, i.e., one produced by the stator current and the other produced by the rotor-induced current will be aligned to each other. The axis of the two magnetic fields will be along the horizontal axis. Since there is no angle of non-alignment between the two magnetic fields, no torque will be developed and hence there will be no rotation of the rotor. **The single-phase induction motor as such will not be self-starting.**



Single-phase induction motor without any starting winding

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Single Phase Induction Motor

According to **double revolving field theory** an alternating magnetic field can be considered equivalent to two revolving fields of constant magnitude (half the magnitude of the alternating field) rotating in opposite directions at synchronous speed (speed of the revolution of the magnetic field in the stator winding of the motor). Thus, an alternating field can be seen as a resultant of two revolving magnetic fields which will have an effect on the rotor which is placed inside that alternating field. The component rotating fields will develop torque on the rotor due to the induction effect and the rotor will rotate on the basis of the resultant torque developed due to these two revolving fields.

This means that when ac is applied, two half magnitude fields are produced with equal magnitudes but revolving in opposite directions. The rotor, however, rotates in one direction due to the effect of resultant of the two torques developed in opposite directions. So, now there is a current flowing in the stator and magnetic field revolving on the rotor, thus Faraday's law of electromagnetic induction acts on the rotor. According to this law, the revolving magnetic fields produce electricity in the rotor which generates a force that can rotate the rotor. These motors are typically used in **fans, refrigerators, air conditioners, vacuum cleaners, washing machines, centrifugal pumps, tools, small farming appliances, blowers** etc.

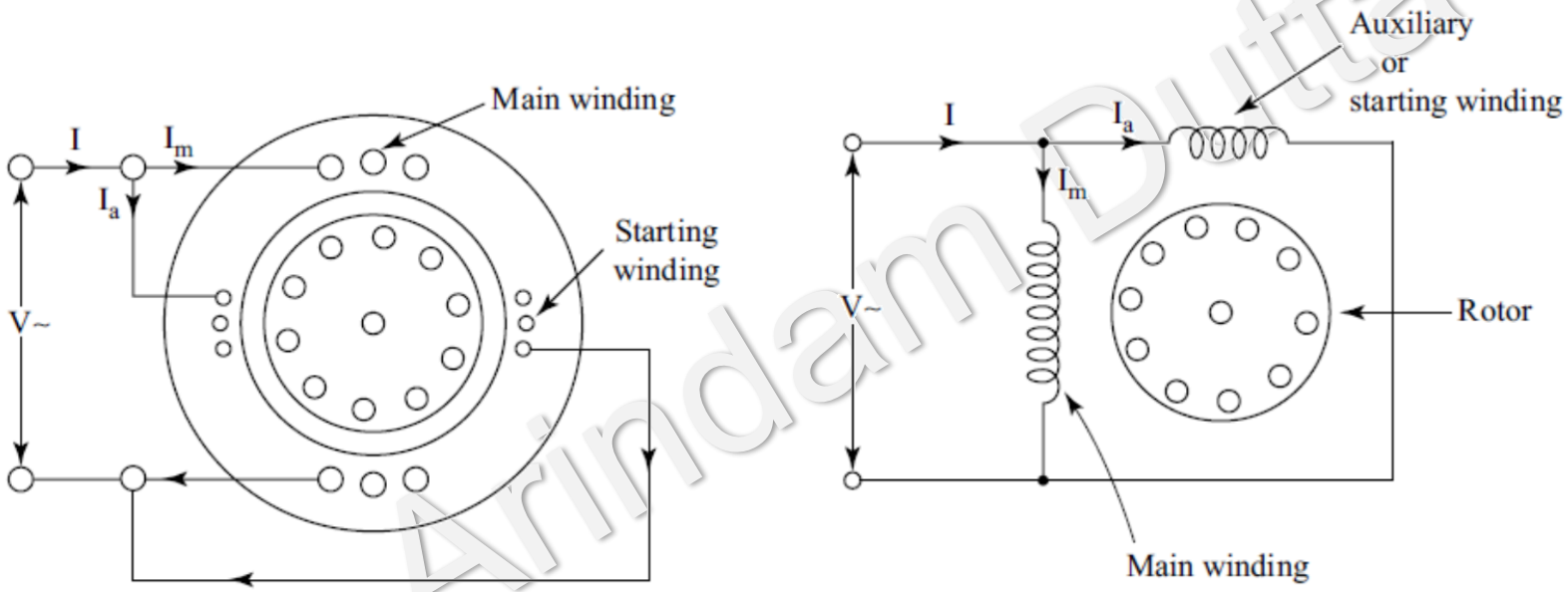
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Two common types:

1. Split phase: Single-phase induction motors are made self-starting by using an additional winding or starting winding placed at an angle of 90° with the main winding in the stator. When the windings are connected in parallel to the source of single-phase ac supply, the starting winding carries a current I_a while the main winding carries a current I_m. The starting winding is made highly resistive while the main winding highly inductive. As a result, a weak revolving field is produced

Single Phase Induction Motor

which starts the motor. When the motor speed reaches about 80% of the synchronous speed, the motor operates as a single-phase induction motor and continues to accelerate till it reaches the normal speed. The normal speed of the motor is less than the synchronous speed and it depends upon the mechanical load on the shaft of the motor.

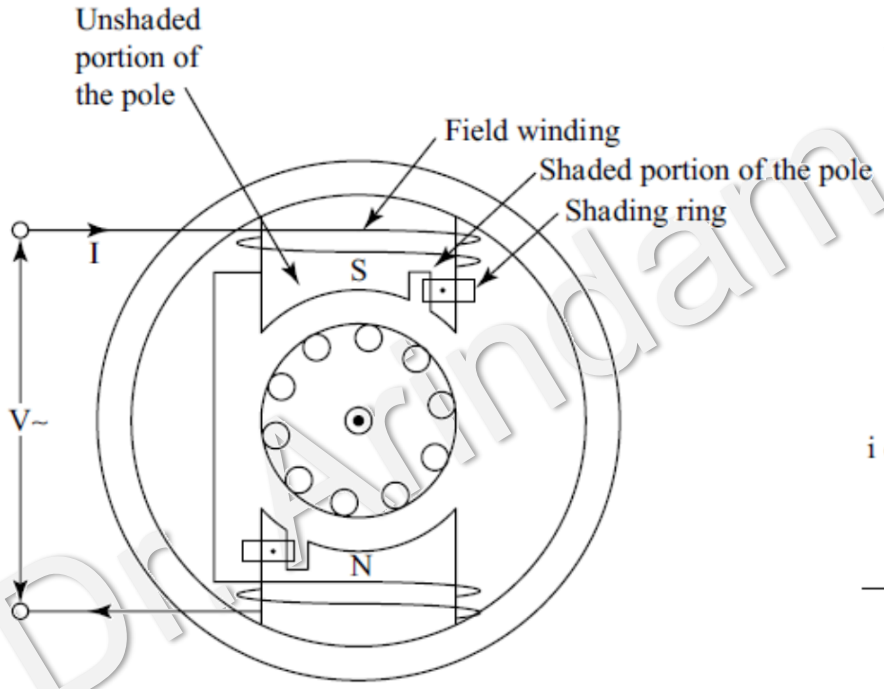


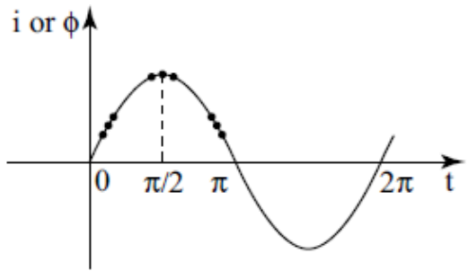
2. Shaded pole: Shaded pole-type single-phase induction motors are provided with shading rings on their poles which are the projected type of poles. The stator of such motors have projected poles like dc machines. The poles are excited by giving single-phase ac supply. Single-turn thick coil in the form of a ring, called the shading ring is fitted on each side of every pole. The portion of the poles where the shading ring is fitted is called the shaded portion, while the other portion is called the unshaded portion. When a single-phase supply is connected across the field windings an alternating current will flow and produce an alternating flux and an EMF will be induced in the rotor conductors. Since the rotor conductors are

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Single Phase Induction Motor

connected together, current will flow through them. In every half cycle of current flow through the field winding, the magnetic neutral axis shifts from the unshaded portion of the pole to the shaded portion. This shift of magnetic axis, creates a torque on the rotor and the rotor starts rotating. Once the rotor starts rotating, it picks up speed and attains its full speed.





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Three Phase Induction Motor

Here, only three-phase supply is required for the stator. No supply is to be provided to the rotor. The rotor is energized due to electromagnetic induction. A three-phase induction motor will have three windings placed in stator slots 120° apart connected either in star or in delta formation. Three-phase supply is provided to these three windings. Due to electromagnetic induction, EMF will be induced into the rotor winding, and if the rotor winding is closed, current will flow through the rotor winding. The interaction between the field produced, due to current flow in the stator windings, when fed from a three-phase supply, and the current-carrying rotor conductors will produce a torque which will rotate the rotor. This is the basic principle of an induction motor.

Constructional details: The main parts are the stator and the rotor. The stator is a hollow cylindrical structure while the rotor is a solid cylindrical body which is placed inside the stator supported at the two ends by two end shields. A small air gap is maintained between the stator and the rotor so that the rotor can rotate freely. The rotor shaft is held at the two ends by two bearings so that the frictional loss is minimum.

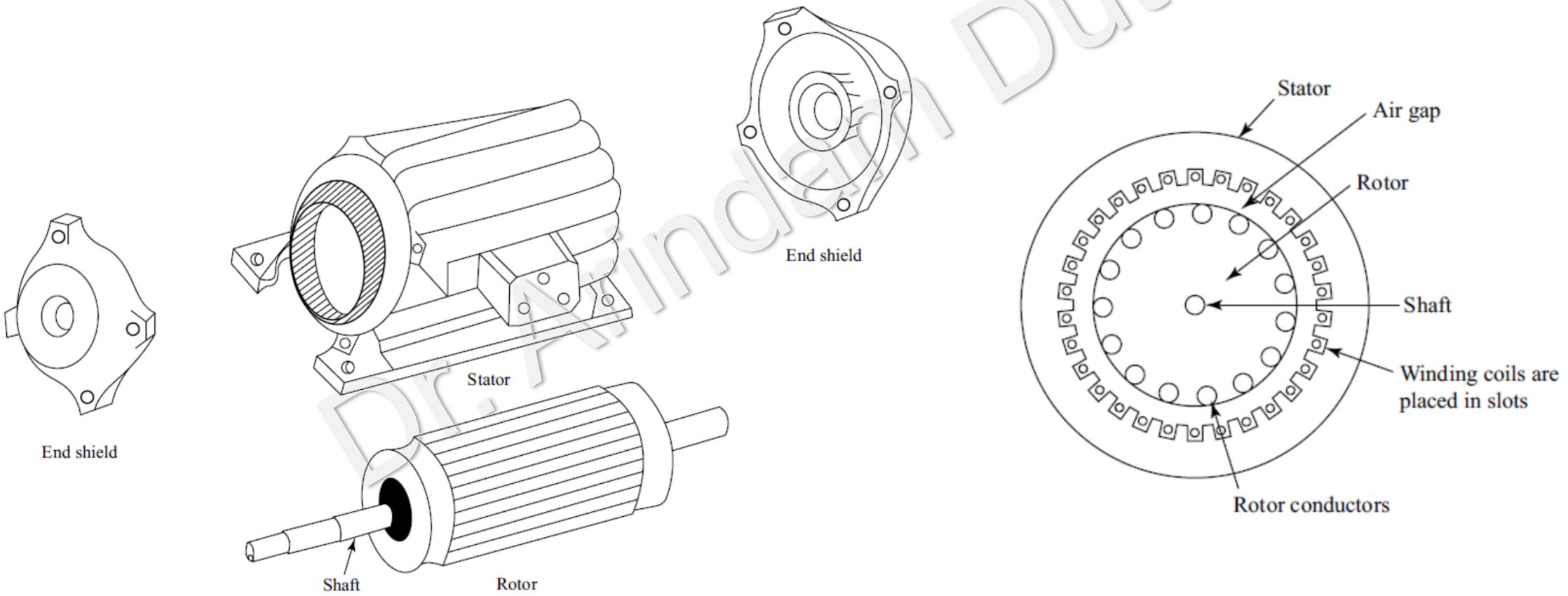
The rotor is placed inside the stator. When the end shields are fitted to the stator from two sides with the rotor shaft passing through the bearings, the rotor will rest on the bearings and the rotor will remain separated from the stator by a small air gap. The three-phase windings are made on the stator. The windings, made of a number of coils, are placed in slots in the stator. Three-phase winding consists of three identical windings separated from each other by 120° in space. Phase supply can be connected to the three open terminals R, Y, B.

The rotor of an induction motor is of two types, namely, squirrel-cage type or slip-ring type. In squirrel-cage type, the rotor winding is made of bars inserted in slots made on the rotor surface. The bars are pushed into the slots and are connected from both sides through conducting rings.

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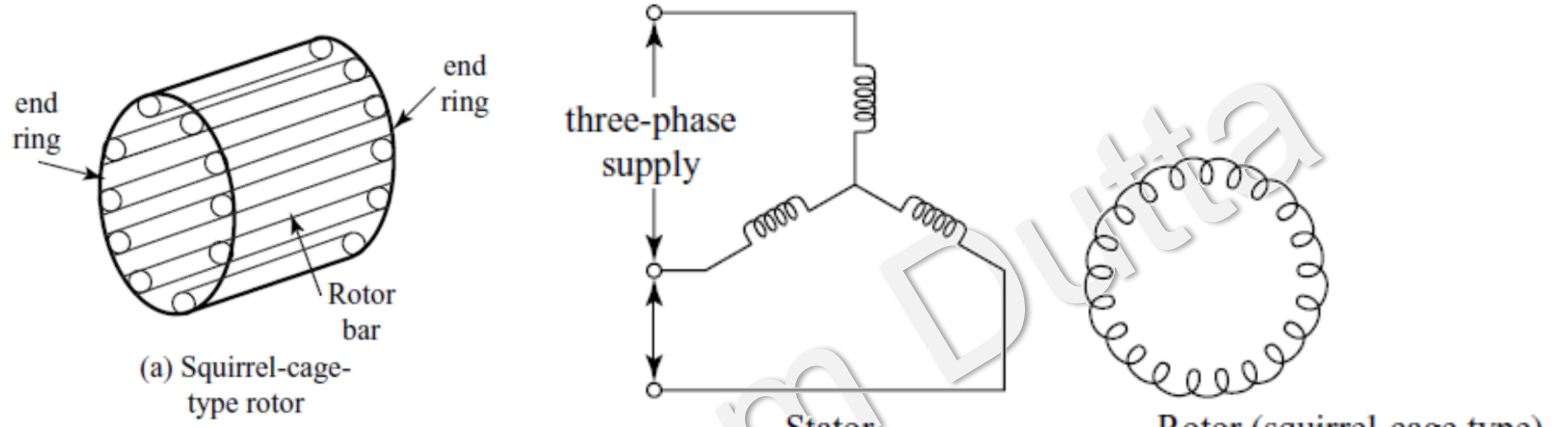
Three Phase Induction Motor

Extra resistance can be connected in the rotor circuit through brush and slip-ring arrangement. The rotor along with the slip rings mounted on its shaft is free to rotate, while brushes and the extra resistance are stationary. By connecting extra resistance in the rotor circuit during starting, very high starting torque can be developed in slip-ring motors. For squirrel-cage rotor, it is not possible to add any extra resistance in the rotor as the circuit is closed by itself permanently and no terminals are brought out.

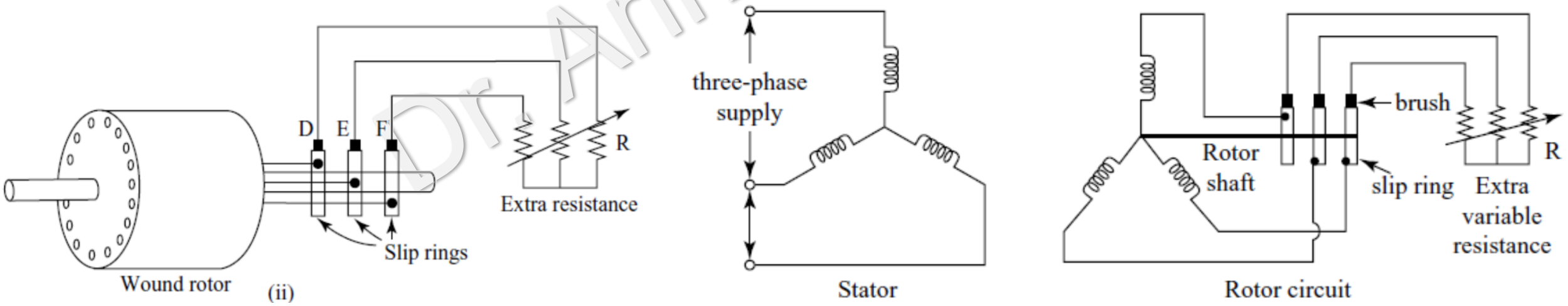


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Three Phase Induction Motor



Squirrel cage type stator and rotor



Slip ring type stator and rotor

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Three Phase Induction Motor

Working principle: In a three-phase induction motor, the stator is wound with a three-phase winding for P number of poles. The poles for which the winding is made could be 2, 4, 6, 8, … etc. The rotor which is placed inside the stator is either squirrel-cage type or slip-ring type. In both cases, current flowing through the three-phase stator winding produces a rotating magnetic field which will be rotating at a speed, N_s where

$$N_s = \frac{120f}{P}$$

For a 50 Hz supply and P = 2, N_s is 3000 rpm. The rotating field will be rotating continuously at a very high speed. The rotating flux will cut the stationary rotor windings at that speed. Due to this cutting of flux, EMF will be induced in the rotor winding. As the rotor circuit has been made to be a closed winding, current will flow through the rotor-winding conductors. Thus, the rotor circuit gets excited due to electromagnetic induction effect. Because of interaction between the current-carrying rotor conductors and the rotating magnetic field, torque will be developed in the rotor, which will rotate the rotor in the same direction as the rotating magnetic field at a speed N_r.

The rotor will attain a speed N_r which is somewhat less than the speed of the rotating magnetic field, N_s. Although the rotor will try to attain a speed of N_s, it will never be able to attain that speed, because if it does, there will be no relative velocity between the rotating field and the speed of the rotor, no EMF induced in the rotor, no current flow in the rotor conductors, no torque developed, and no rotation of the rotor. **That is why an induction motor cannot run at synchronous speed, N_s** as it is to be excited by electromagnetic induction, which is possible only if there exists a relative velocity between the rotating magnetic field and the rotor.

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Three Phase Induction Motor

Slip: The difference between the speed of the rotating magnetic field, N_s and the rotor speed N_r is the slip S. Slip is usually expressed as the percentage of N_s, thus,

$$S = \frac{N_s - N_r}{N_s}$$

$$SN_s = N_s - N_r \text{ or, } N_r = (1 - S)\, N_s$$

Percentage slip,

$$= \frac{N_s - N_r}{N_s} \times 100$$

Slip of a three-phase induction motor is generally 3 to 4 per cent. For example, when a 400 V, 3-ph, 50 Hz supply is connected to a four-pole three-phase induction motor, the speed of the rotating field will be 1500 rpm. The rotor will rotate at a speed less than 1500 rpm, may be, say 1440 rpm. In such a case slip S is

$$S = \frac{(1500 - 1440) \times 100}{1500}$$

$$= 4 \text{ per cent}$$

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EEE1001
ELECTRIC CIRCUITS AND SYSTEMS

Module 3 – Lecture 4

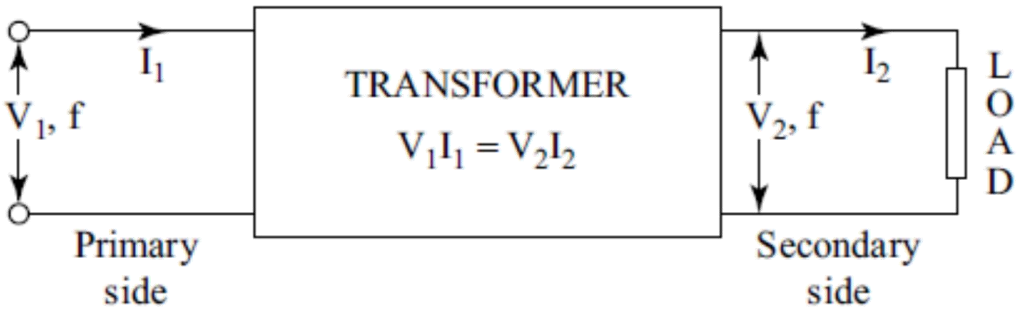
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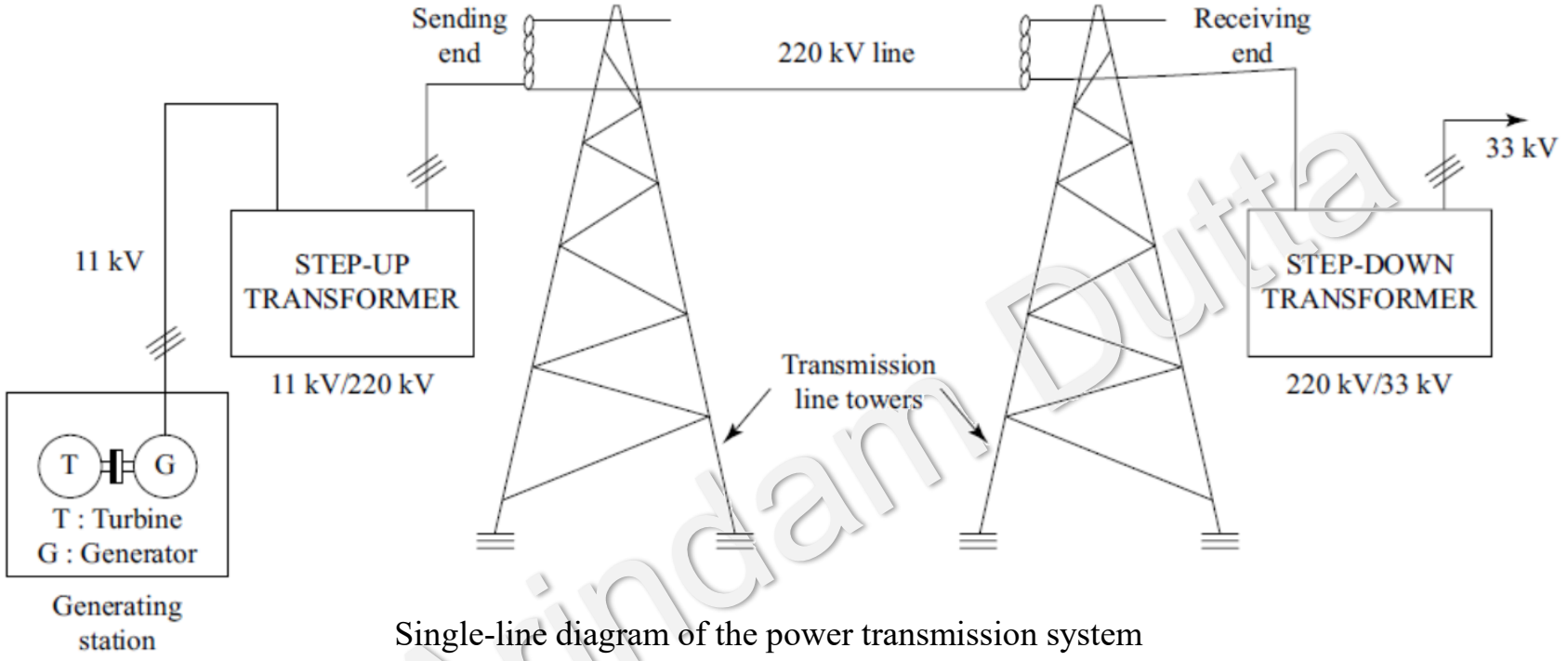
Transformers

- The generation voltage in power stations is 11,000 V or 11 kV
- Electricity is transmitted from the place of generation to the place of its use through electrical transmission lines
- The voltage level of the power to be transmitted is raised to higher values, say 220 kV to reduce the cost of transmission
- It is again lowered to 230 V or at 400 V before supplying to consumers, for safety reasons
- **Transformers** are used at both the ends of the transmission lines to raise or lower the voltage level of electrical power
- When voltage is raised from lower level to higher level, the device used is called a **step-up transformer**
- When voltage is lowered from a high level to a low level, the transformer used is called a **step-down transformer**
- The frequency of the input voltage is same as that of the frequency of the output voltage

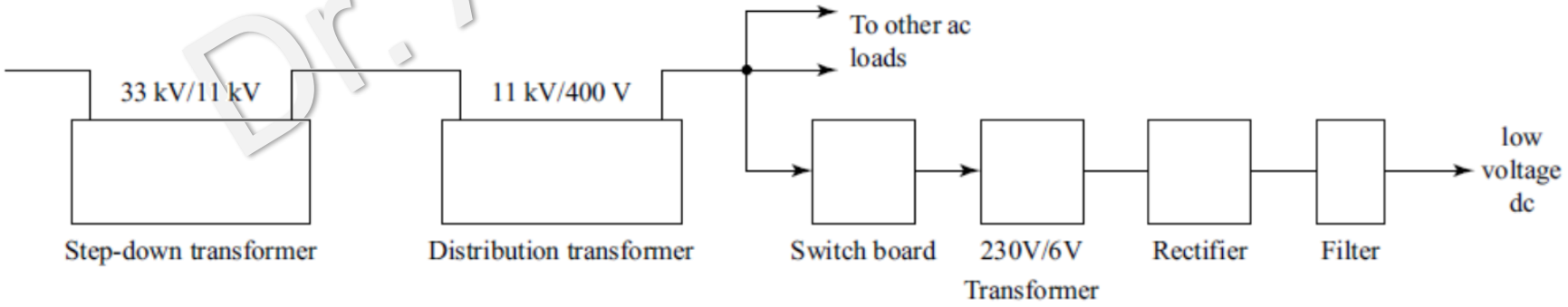


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Transformers



Single-line diagram of the power transmission system



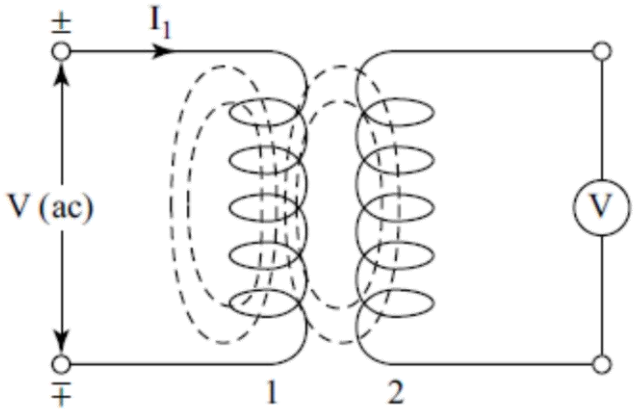
Block-diagram representation of the distribution of electricity for end use

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Basic Working Principle

Transformers work on the principle of electromagnetic induction, that is, emf is induced in a coil due to the rate of change of flux linkage by it. In a transformer, two coils, which are also called windings wound on an iron core, are used. Coil 1 is called the primary winding and coil 2 is called the secondary winding. These windings are made on an iron core which is made of a magnetic material. The magnetic material permits easy establishment of the flux through the core, and hence through the windings. Since it is necessary to produce more flux by using small current, the reluctance of the flux path must be low. Iron has high permeability, i.e., low reluctance. Use of iron as the core material for the windings improves the magnetic coupling between the windings, which is essential for the transfer of power from one circuit to the other efficiently.

By changing the ratio of the number of turns of the coils the magnitude of the induced EMF in the second coil can be changed. The frequency of the induced EMF in the two coils will be the same as that of the frequency of power supply to the first coil. Now, if an electrical load is connected across the second coil, power will be delivered to the load. Thus, when electrical supply is connected to the primary circuit or the winding, power gets transferred to the second circuit via the magnetic circuit. Thus, transformers are based on the principle of magnetic coupling of two coils.

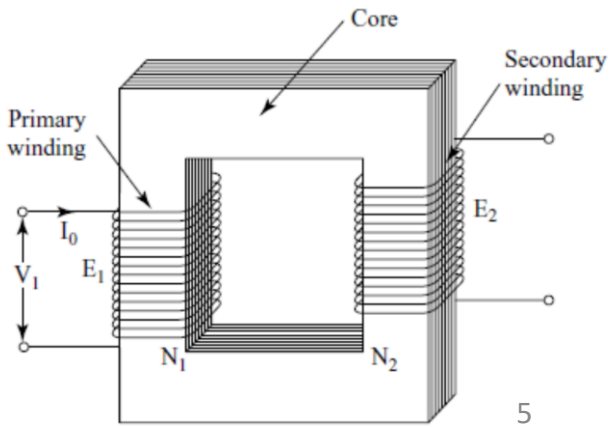


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Constructional Details

The transformer consists of a core made up of a magnetic material around which two coils are placed. One coil is connected to supply voltage. This coil is called the primary winding. The other coil is called the secondary winding. Supply is taken from the secondary winding by connecting any electrical load like fan, tube light, electrical motors, etc. The transformer core is made up of thin sheets called laminations. This is done to reduce power loss in the core due to eddy current flowing in the core and producing undesirable heating of the core as well as of the windings which are wound on the core. The laminated silicon steel sheets are cut into proper size from a big sheet and are placed one above the other to form the core of required width and cross section. These are tightly fastened to form the core. Hysteresis loss occurs due to orientation of the magnetic dipoles of the magnetic material when the material is subjected to the alternating magnetic field. When some percentage of silicon is added to steel, hysteresis loss in the core gets reduced.

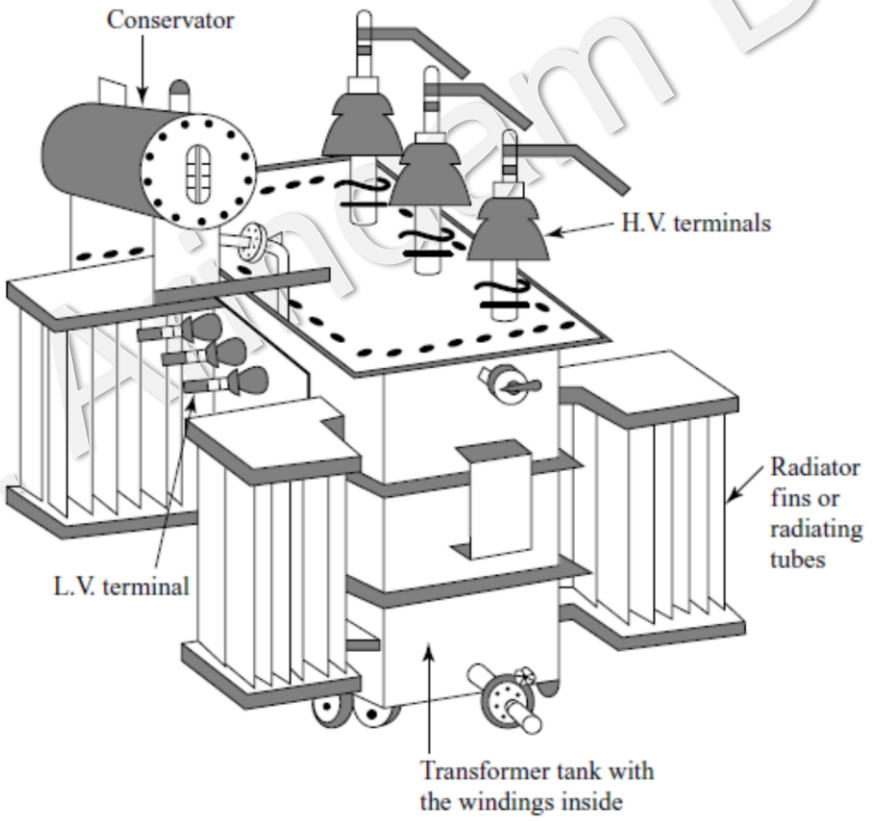
The windings of the transformer are made up of insulated copper wires. The cross-sectional area of the winding wires will depend upon the requirement of current-carrying capacity, and the number of turns are calculated according to the voltage ratio of primary and secondary windings. The core and the winding assembly are placed inside a tank filled with transformer oil for the purpose of providing insulation to the windings and also for cooling purpose. Transformer oil used is mineral oil having high dielectric strength. The tank is provided with radiating tubes so that heated oil gets circulated through the tubes and heat produced in the transformer is radiated to the atmosphere through the oil circulating from the tank through the radiating tubes.



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Constructional Details

Transformers are manufactured as single-phase transformers and as three-phase transformers. In three-phase transformers three separate windings are made for both primary and secondary sides. The windings are connected either in star or in delta. Terminal connections are brought out through low voltage (L.V.) terminals and high voltage (H.V.) terminals. A conservator tank fitted with a breather is placed above the tank. The conservator is connected to the transformer tank with a pipe and carries transformer oil. The empty space above the level of oil in the conservator is provided to allow the expansion of oil in the tank due to heating and for the removal of gas formed.

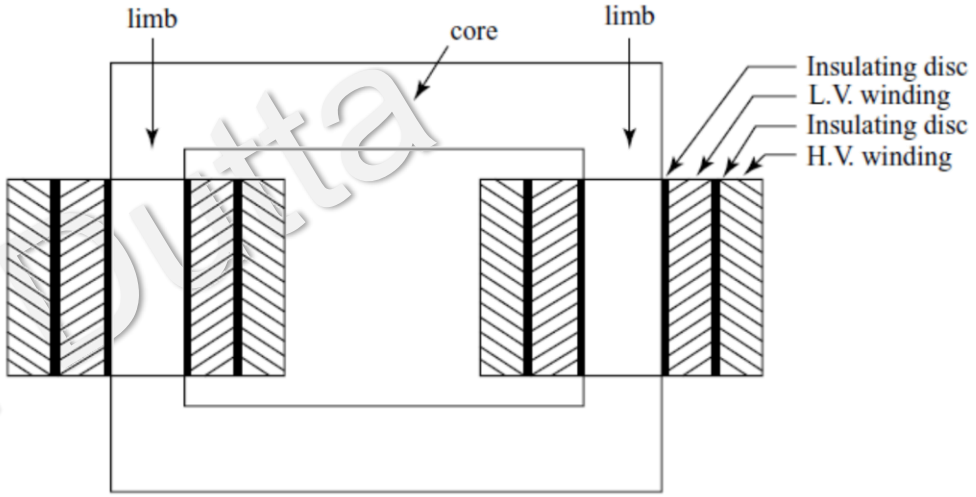


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Core and Shell Types

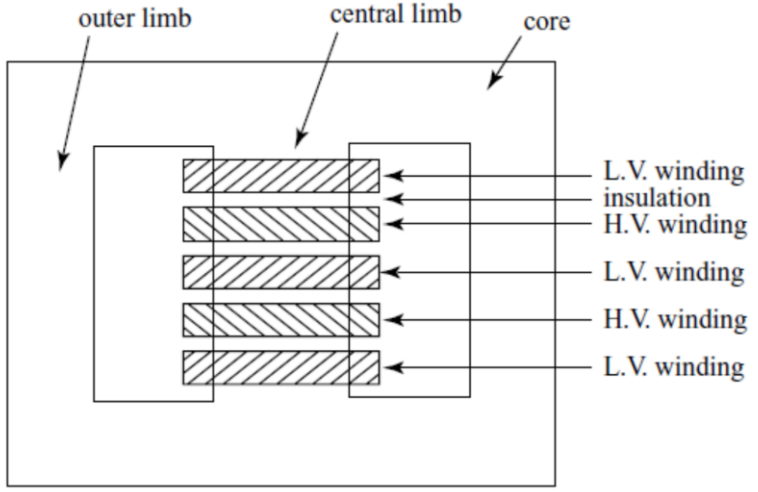
There are two types of core constructions: core type and shell type.

In **core-type construction**, the primary and secondary windings are placed around the limbs of the transformer core. First, the low-voltage winding is placed around the limbs. Over the low-voltage windings are placed the high-voltage windings. The high-voltage winding is placed somewhat away from the core so as to reduce the insulation problem. The windings are insulated from the core through insulating cylindrical disc made of insulating material. The windings are made up of insulated copper wires in two sections or parts and are connected after they are placed in positions to form primary and secondary windings, respectively.



Core type

In a **shell-type construction**, the windings are placed in the central limb. The windings are wound in the form of a number of circular discs, and are placed one above the other. The extreme two discs on the central limb are low-voltage winding sections. These, l.v. sections and h.v. sections are then connected to form low-voltage and high-voltage windings. The width of the central limb is twice the width of the outer limbs so that the flux density is the same in all the limbs.



Shell type

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EMF Equation

The general equation for the instantaneous value of the EMF induced, e is expressed as $e = -N\frac{d\phi}{dt}$

Primary winding has N_1 turns. The induced EMF in the primary winding E_1 is

$$E_1 = 4.44\,\phi_m\,f\,N_1\,V$$

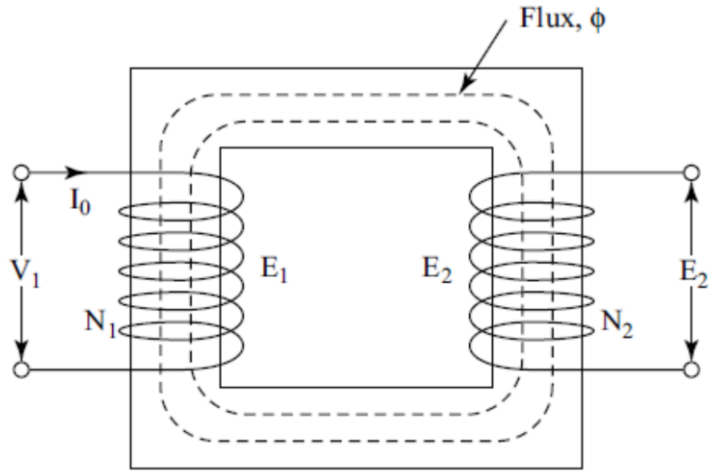
Where, φ_m is the peak value of the flux and f is the frequency. Since the same flux producing E_1 also links the secondary winding having N_2 number of turns, the induced EMF in the secondary winding, E_2 is

$$E_2 = 4.44\,\phi_m\,f\,N_2\,V$$

Dividing,

$$\frac{E_1}{E_2} = \frac{N_1}{N_2}$$

$$\frac{E_2}{E_1} = \frac{N_2}{N_1} = K$$



K is called the ratio of voltage transformation. For a step-up transformer, $N_2 > N_1$, and hence, the value of K is more than 1. For a step-down transformer, N_2 is less than N_1, and hence $K < 1$.

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Problem

Q: A transformer has 1000 turns on its primary and 500 turns on the secondary. When a voltage, V of frequency 50 Hz is connected across the primary winding a maximum flux of 2×10^{-3} Wb is produced in the core which links both the windings. Calculate the value of the EMF induced in the two windings.

Let E_1 and E_2 be the EMFs induced in primary and secondary windings, respectively. Here, $N_1 = 1000$ and $N_2 = 500$.

$$E_1 = 4.44 \phi_m f N_1$$
$$= 4.44 \times 2 \times 10^{-3} \times 50 \times 1000.$$
$$= 444 \text{ V}$$
$$E_2 = 4.44 \phi_m f N_2$$
$$= 4.44 \times 2 \times 10^{-3} \times 50 \times 500$$
$$= 222 \text{ V}$$

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Problem

Q: A transformer has 900 turns on its primary winding and 300 turns on its secondary. A voltage of 230 V at 50 Hz is connected across its primary winding. The cross-sectional area of the core is 64 cm^2. Calculate the magnitude of the induced EMF in the secondary winding. Also calculate the value of maximum flux density in the core.

Given $V_1 = 230$ V. The induced EMF in the primary windings is E_1. Typically, E_1 is slightly less than V_1 because there will be some voltage drop in the winding. Neglecting the voltage drop in the winding, we can write $V_1 = E_1$.

$$E_1 = 4.44\,\phi_m\,f\,N_1$$

$$E_2 = 4.44\,\phi_m\,f\,N_2$$

$$\frac{E_1}{E_2} = \frac{N_1}{N_2}$$

$$E_2 = E_1\left(\frac{N_2}{N_1}\right) = 230\left(\frac{300}{900}\right) = 76.7\ \text{V}$$

$$E_1 = 4.44\,\phi_m\,f\,N_1$$

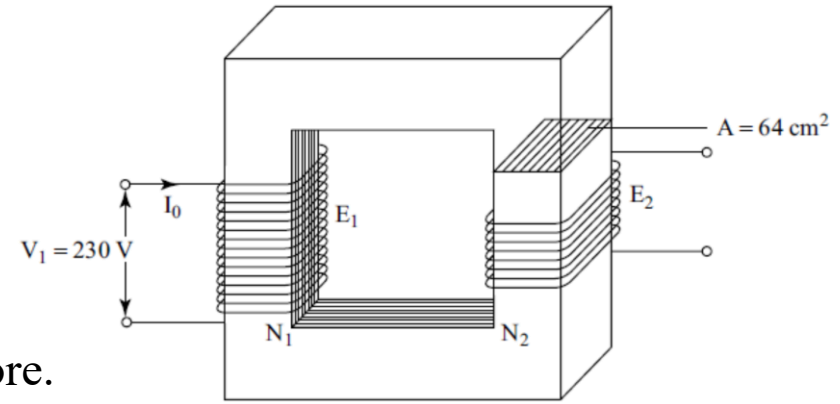
$$= 4.44\,B_m\,A\,f\,N_1$$

where B_m is the maximum flux density and A is the cross-sectional area of the core.

Thus, $$230 = 4.44\,B_m \times \frac{64}{10^4} \times 50 \times 900$$

$$B_m = \frac{230 \times 10^4}{4.44 \times 64 \times 45000} = \frac{2300}{4.44 \times 64 \times 45}$$

$$B_m = 0.18\ \text{Wb/m}^2$$



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